Governance Concentration Beyond Token Allocation: An Institutional Design Analysis of DePIN and DeFi

Zach Zukowski

Tokenization Systems

zach@tokenization.systems

ORCID: 0009-0006-3642-2450

April 2026

Abstract

Initial token allocation design does not predict governance concentration. A 40-protocol cross-section spanning Decentralized Physical Infrastructure Networks (DePIN), DeFi, infrastructure, and social token categories documents that the percentage of tokens allocated to team and investors at launch is uninformative about post-distribution governance concentration (Pearson $r = 0.18$, $p = 0.28$, $N = 37$): protocols with generous community distributions exhibit concentration patterns similar to those with heavy insider allocations. Hyperliquid, the only protocol in the sample with zero venture capital and a 31% community airdrop, shows the lowest DeFi-subsample Herfindahl-Hirschman Index (HHI) of 0.005, but the broader null holds across allocation designs. Evidence is consistent with post-distribution market dynamics shaping concentration more than initial structural design.

Four supporting findings sharpen the picture. First, protocols with more insider wallets in their top-holder sets exhibit higher concentration among non-insider holders (Spearman $\rho = 0.54$, $p = 0.001$, $N = 34$), indicating that insider-heavy protocols develop concentrated governance ecosystems, not merely concentrated insider positions. Second, sector membership predicts concentration: DePIN protocols are more concentrated than DeFi (Mann-Whitney $p = 0.014$, Cohen's $d = 1.03$), and the effect survives multivariate OLS adjustment across three nested specifications adding protocol age, log fully diluted valuation (FDV), and initial insider allocation as successive controls (sector coefficient $p < 0.05$ in Models 1 and 2, borderline at $p = 0.050$ in Model 3 with the full control set; adjusted R^2 0.14 to 0.17; Table 6). Third, on-chain subsidy intensity correlates with concentration in levels ($r = 0.57$, $p = 0.008$, $N = 20$) but entirely through Livepeer (88.5x subsidy): excluding Livepeer, the correlation is not significant ($r = 0.11$, $p = 0.65$). Fourth, delegation amplifies concentration unevenly: voting-power HHI reaches 3 to 6 times holding HHI in DePIN, while two infrastructure protocols diverge (Optimism 0.79x, Arbitrum 4.3x), consistent with institutional design at the operator and router level shaping governance outcomes more than sector classification alone.

These findings rest on a measurement distinction often conflated in the literature. Token inequality (Gini, dispersion across all holders) and governance concentration (HHI, sum of squared shares across top holders) capture distinct properties of the same holder set. Inequality is severe in every protocol (Gini 0.52 to 0.99), while concentration varies across two orders of magnitude (HHI 0.005 to 0.199); their moderate correlation (r = 0.51) shows that inequality metrics cannot substitute for direct concentration measurement.

JEL Codes: C21, D02, D47, D63, D71, D72, G34, L22, P48

Keywords: tokenomics, institutional design, governance concentration, DAO governance, DePIN, delegation amplification, Herfindahl-Hirschman Index, political philosophy of institutions, polycentric governance, exclusion methodology

1 Introduction

1.1 Motivation

Three protocols with near-identical subsidy ratios exhibit governance HHI spanning 0.020 to 0.171: financial sustainability is neither necessary nor sufficient for governance decentralization (Section 3.4). Governance concentration is a secondary market outcome, shaped by post-distribution trading and accumulation dynamics that neither allocation design nor protocol financial characteristics predict.

Token-based governance is the default coordination mechanism for billions of dollars in decentralized physical infrastructure (DePIN) and decentralized finance (DeFi). But what determines whether governance power concentrates or distributes? Three candidate explanations dominate practitioner discourse: insider token allocation concentrates power; subsidy dependence (high token emissions relative to revenue) entrenches early holders; and younger protocols have not yet had time to decentralize. This paper tests all three.

We construct a 40-protocol governance concentration dataset spanning DePIN and DeFi, develop a Blockscout-verified (a blockchain contract verification tool) exclusion methodology to remove 69 protocol-controlled addresses (staking contracts, exchange custodians, vesting locks, and protocol treasuries) from HHI computation, and test every available cross-sectional predictor of concentration using both parametric and rank-based methods.

Four findings emerge. First, naive token holder lists include addresses that cannot vote: staking contracts, exchange custodians, vesting locks, and protocol treasuries. Across 20 protocols, 69 such addresses were included in prior concentration measurements, inflating affected protocols' HHI by up to 7x. Second, current insider wallet retention predicts governance concentration (Spearman rho = 0.48, $p = 0.003$, $N = 37$), while initial allocation design does not (Section 3.4). Third, DePIN governance concentration is higher than DeFi after exclusion correction (Mann-Whitney $p = 0.014$, Cohen's $d = 1.03$), and the finding is robust to single-protocol exclusion (significant in all 30 of 30 leave-one-out iterations). Fourth, delegation amplifies concentration unevenly: DePIN protocols show voting-power HHI 3 to 6 times their holding HHI, while the two infrastructure protocols sampled show heterogeneous outcomes (Optimism 0.79 times, Arbitrum 4.3 times), suggesting that institutional design within the delegate program rather than sector category determines the direction of delegation effects.

Recent scoping work by Alshater (2026) systematically maps the DePIN tokenomics design space, identifying the burn-and-mint equilibrium, governance-calibrated issuance, and provider staking as the dominant economic primitives across leading deployments. Alshater explicitly frames decentralization as a spectrum rather than a binary property, echoing analyses of decentralized governance for cyber-physical systems (Nabben et al., 2024), and identifies the lack of rigorous quantitative empirical work on these models as the field's most pressing research gap. The cross-sectional measurement reported here responds to that gap, applying Herfindahl-Hirschman Index analysis to forty protocols spanning DePIN and adjacent categories.

These results extend the governance concentration literature (Barbereau et al., 2023; Feichtinger et al., 2023; Cong et al., 2025) in three directions: an exclusion methodology that corrects a systematic measurement artifact in prior work, a cross-sector null documenting that neither revenue, subsidy intensity, token maturity, nor initial allocation predicts governance concentration, and a normative framework translating five political-philosophical traditions into observable design criteria for tokenized institutions. To evaluate what concentration means for institutional legitimacy, the framework operationalizes constraints from Kant (publicity), Rawls (floor protection), Pettit (non-domination), Ostrom (polycentricity), and Hayek (dispersed knowledge) through a scoring rubric that other researchers can apply. This paper is part of a research program examining infrastructure-layer design across stablecoin, tokenized asset, and DePIN domains. The program's unifying thesis: regulation and audit should target the operators and control systems that issue, custody, and route tokens, rather than the tokens themselves.

1.2 Core Thesis: Tokenomics Is Applied Political Philosophy

The central claim:

Tokenomics is applied political philosophy: the encoding of governance, rights, duties, distributive choices, and legitimacy constraints into a rule system with economic incentives and automated enforcement.

In practical terms: every decision about how tokens are distributed, who can vote, what can be changed, and how violations are punished is simultaneously an economic design choice and a political one.

Political philosophy asks: *What makes a rule system just, stable, and worth participating in?* Tokenomics asks: *What rules and incentives produce participation, security, sustainability, and coordination under strategic behavior?* In tokenized systems, these questions converge. Economic design choices are simultaneously political choices: they distribute power and resources, shape inclusion or exclusion, and determine whether participants view outcomes as fair and contestable.

Philosophical principles function here not as commentary but as specification: constraints and requirements that token mechanisms should satisfy. It draws particularly on:

Kantian publicity and right (rules must be publicly avowable; enforcement must respect due process and non-arbitrary coercion),

Rawlsian legitimacy and floor protection (rules should be defensible under public reason; distribution should not sacrifice the least advantaged),

Republican non-domination (participants should not be vulnerable to arbitrary interference; contestability is essential),

Ostromian polycentric governance (multiple centers of rule-making and monitoring improve robustness in complex commons),

Hayekian dispersed knowledge (local signals and edge decision-making outperform centralized tuning in heterogeneous environments),

Capabilities and information ethics (success includes social opportunity and privacy-preserving participation, not only token price and growth).

The resulting framework is explicitly cross-disciplinary: it aims to speak simultaneously to political philosophy, economics/public policy, computer science/security, law, and Science and Technology Studies (STS).

1.3 Research Questions

Three research questions guide the analysis:

RQ1: Can normative political philosophy be systematically translated into evaluable design criteria for tokenized institutions? If so, what does the resulting framework reveal about the legitimacy properties of existing protocols?

RQ2: What does the distribution of governance tokens reveal about concentration of power in tokenized institutions, and how do DePIN protocols compare to established DeFi governance systems?

RQ3: What design hypotheses emerge from the normative framework, and which are supported by the governance concentration evidence? Longitudinal case evidence on Helium's fiscal sustainability and demand concentration is developed in a separate study (Zukowski, 2026a).

1.4 Scope, Assumptions, and Limitations

1.4.1 Scope

The empirical analysis focuses on tokenized systems between 2017 and 2025 across DeFi, infrastructure protocols, DAOs, and DePIN networks. The core analytic target is **institutional design quality** rather than short-term market performance.

1.4.2 Assumptions

Participants are strategic; incentives and governance rules shape behavior.

Legitimacy affects participation persistence and willingness to invest in long-run cooperation.

Public rule systems can be observed and scored with acceptable reliability using well-defined rubrics.

1.4.3 Limitations

Construct validity: translating philosophical traditions into observable governance features carries inherent simplification risks.

Causal inference scope: the analysis is descriptive. A cross-sectional snapshot measures associations between protocol features and governance concentration at one point in time; it cannot isolate which feature causes which outcome. The correlations reported here are interpreted as associations, not as causal mechanisms.

Data limitations: not all outcomes relevant to institutional legitimacy are observable on-chain; capability and welfare dimensions require complementary data beyond the current analysis.

1.5 Paper Outline

Section 2 develops the theoretical framework and design hypotheses and (in Section 2.10) operationalizes the framework through a scoring rubric and governance concentration cross-section design. Section 3 presents the empirical illustration. Section 4 discusses implications, limitations, and future directions. Supplementary materials provide full metric definitions, research instrument templates, and the replication-ready pipeline specification.

2 Materials and Methods

2.1 Overview: From "Token Design" to Institutional Design

The mainstream discourse around tokenomics often treats it as a subset of product pricing or financial engineering (see Voshmgir, 2020, for a representative treatment), supply curves, vesting schedules, inflation, and speculative demand. However, a growing body of scholarship and technical work suggests a broader interpretation: token systems encode rules of participation, allocate rights and duties, and implement enforcement and dispute resolution through mechanisms that are both transparent and automatically executed by software.
In this view, tokenomics is not merely an economic optimization problem but an instance of institutional design (and therefore a normative problem) in which governance, legitimacy, and distributive consequences are intrinsic rather than externalities.

Token systems function as institutions: rule-governed systems with constitutional constraints, fiscal policy, property-like rights, and enforcement procedures. This framing parallels classic traditions in constitutional political economy (e.g., Buchanan and Tullock (1962)) and commons governance (Ostrom), while also drawing from political philosophy's theories of legitimacy, justice, and domination. A key theme is that crypto networks differ from Web2 platforms not because they "lack governance," but because they encode governance into public, inspectable mechanisms and often bind governance to ownership-like tokens.

2.2 Tokenomics and Cryptoeconomic Design

2.2.1 Tokenomics as a Mechanism Design Problem

Tokenomics can be understood as a form of mechanism design (Hurwicz, 1973; Myerson, 1981): the "reverse engineering" of game theory in which a designer specifies rules and incentives to induce desired behavior among strategic agents (users, validators, operators, token holders). The mechanism design literature provides foundational formal tools for reasoning about incentive compatibility, equilibrium selection, and adverse selection. These tools remain highly relevant when tokenized systems must function under manipulation pressure and incomplete information. The Nobel Prize's summary of mechanism design theory emphasizes incentive-compatible direct mechanisms and the revelation principle as foundational concepts, offering a bridge between classical economics and modern crypto-economic system design.

Token systems, however, often introduce additional constraints absent from classical mechanism design (Roughgarden, 2021) settings: open entry, pseudonymity, composability, and adversarial execution environments. These constraints place a premium on robust incentive design, fraud resistance, and governance structures that can adapt without becoming arbitrary or captured.

2.2.2 Transaction Fee Mechanisms and "Sinks" as Economic Policy

Some of the clearest illustrations of tokenomics-as-policy appear in fee mechanism design. Ethereum's EIP-1559 (Buterin et al., 2019), for instance, introduced a base fee adjusted by network demand and burned rather than paid to miners/validators, creating a direct "sink" linked to usage and shaping the economic properties of the asset. Tim Roughgarden's work on transaction fee mechanism design treats EIP-1559 not as marketing but as mechanism engineering: explicitly analyzing goals and trade-offs, including alternative designs such as smoothing and revenue-forwarding.

Subsequent research (Roughgarden, 2020) highlights that even well-designed sinks can introduce new strategic behaviors (e.g., manipulation or minority attacks on base fee dynamics), underscoring that tokenomics is not "set-and-forget" but a governance problem requiring transparency and monitoring.

These studies motivate a broader design intuition central to this paper: token mechanics that tie usage to scarcity (through burns, locks, buybacks, or fee routing) represent a form of fiscal policy embedded in code (cf. Werner et al., 2022).

The leading formal model of token adoption (Cong, Li, & Wang, 2021) demonstrates that equilibrium token price is determined by aggregating heterogeneous users' transactional demand rather than by initial supply distribution, implying that post-distribution dynamics dominate design-time allocation choices. This theoretical result anticipates the allocation design null documented in Section 3.4.

2.3 Incentives, Liquidity Mining, and the "Mercenary Capital" Debate

A major wave of Web3 growth (notably in DeFi) was catalyzed by token distribution mechanisms: initial coin offerings (Howell, Niessner, & Yermack, 2020), liquidity mining, staking rewards (Schär, 2021), and user rebates, designed to bootstrap two-sided markets. A key controversy is whether these incentives generate durable adoption or merely transient participation driven by extrinsic rewards.

Empirical and conceptual analyses increasingly emphasize the risk of "mercenary capital" (Schär, 2021): incentives attract liquidity or usage temporarily, but participation exits when emissions decline, leaving weak organic demand. While the quality of evidence varies across sources, the consistent theoretical point is that token incentives can create short-run growth while undermining long-run sustainability unless paired with credible sinks, retention design, and governance discipline. This paper integrates that debate into a more general claim: **incentives are political economy**, not simply growth marketing, because they distribute resources, create winners and losers, and can entrench power.

Academic work on liquidity mining (Schär, 2021; Roughgarden, 2021) explicitly treats it as an innovative mechanism that changes investor behavior and protocol dynamics, suggesting a need for systematic evaluation rather than anecdotal interpretation.

2.4 DAO Governance: Concentration, Delegation, and the "Decentralization Illusion"

The legitimacy of tokenized governance depends not only on formal voting rules but also on power distribution and participation (Aramonte et al., 2021; Nadler & Schär, 2020) structure. A major challenge is that token-based governance can produce concentrated influence (Barbereau et al., 2023) - "whales," delegates, or insiders, leading to outcomes that are formally decentralized but substantively centralized.

The Bank for International Settlements (BIS) argues (Aramonte et al., 2021) that DeFi contains a "decentralization illusion," emphasizingthat protocols marketed as decentralized frequently concentrate decision-making power among a small number of token holders, developers, or validators. This critique is significant for this paper because it links governance concentration to systemic risks and questions the legitimacy of "decentralized" claims.

Gersbach, Schneider, and Shahkar (2024) analyze vote delegation and find it can meaningfully change outcomes under highly unbalanced voting rights but has negligible impact under balanced systems, suggesting delegation is not a universal solution to legitimacy problems. Weidener et al. (2025) similarly maps delegation's implementation and challenges, reinforcing that delegation can either mitigate or entrench concentration depending on design. Hall and Miyazaki (2024) analyze liquid democracy across 18 DAOs, finding that only 17% of tokens are delegated yet the top five delegates routinely capture over 50% of effectively voted weight, with high-SSPI delegates more likely to signal early and cascade alignment; the 3 to 6 times delegation amplification ratios documented in Section 3.5 are consistent with this clumping pattern. Esposito, Tse, and Goh (2025) evaluate quadratic voting and delegated governance against Ostrom's CPR framework, documenting that tokenization alone is insufficient for inclusive governance.

Most directly relevant, Cong et al. (2025) provide stylized facts and analysis suggesting persistent concentration in DAOs and governance token trading dynamics, reinforcing the need for institutional safeguards rather than assuming "token voting" equals democracy. Panel evidence from DAO-quarter data documents that effective governance control concentrates as proposal volume scales, consistent with monitoring costs outpacing broad-based participation (Tchuente, 2025); the cross-sectional allocation null documented here ($r = 0.18$, $p = 0.28$) is compatible with this scale-driven concentration mechanism operating independently of initial token distribution.

Calzada (2025) corroborates these patterns ethnographically across seven Web3 ecosystems, documenting whale dominance and voter turnout below 10% in major DAOs.

The governance concentration patterns documented in Section 3 are consistent with Williamson's (1999) "probity hazard": token holders can comply with smart contract rules while extracting value that contradicts the protocol's institutional mission, a risk that intensifies as governance power concentrates in fewer addresses.

Voting power inequality in Compound, Uniswap, and ENS exceeds wealth inequality in real-world economies, withvoting power more unequally distributed than income in any country (Fritsch, Müller, & Wattenhofer, 2024), a pattern consistent with the delegation amplification documented in Section 3.5; notably, these concentrated delegates rarely overturn community-preferred outcomes, suggesting that concentration creates domination risk rather than routine domination. Across 10,639 proposals in 151 DAOs, three or fewer entities exert effective control over most governance decisions (Appel & Grennan, 2023), consistent with the insider dominance this paper documents at the token-holder level.

Together, these literatures (see also Feichtinger et al., 2023; Rikken et al., 2023) motivate two design imperatives advanced in later chapters: (i) governance systems require **contestability and due process** beyond raw voting, and (ii) legitimacy depends on measurable concentration constraints and multi-constituency representation.

Han, Lee, and Li (2023) formalize the governance consequences of ownership concentration, showing that whale dominance reduces platform growth but that staking mechanisms and token illiquidity can mitigate it. The finding that ownership concentration is structurally harmful, not merely descriptive, motivates the normative framework developed here: if concentration damages governance quality, the political-philosophical traditions that specify what good governance requires become directly relevant to token system design. Han, Lee, and Li (2024) provide a comprehensive review of DAO governance mechanisms and the emerging agency problem between large and small token holders.

2.4.1 Cooperatives as the Nearest Institutional Ancestor

Among established institutional forms, the cooperative bears the closest structural resemblance to the tokenized protocol. Both organize collective production without a single owner who captures surplus; both distribute governance rights to participants rather than to external shareholders; both face the challenge of sustaining voluntary participation under conditions where exit is available. The cooperative literature (Hansmann, 1996; Birchall, 2011) identifies these properties as the defining features of member-owned enterprise, distinguishing cooperatives from investor-owned firms. The International Cooperative Alliance's seven cooperative principles (ICA, 1995), including voluntary membership, democratic member control, and concern for community, map onto legitimacy constraints derived from political philosophy.

Five structural differences separate protocols from cooperatives. First, cooperatives require application and capital contribution; protocols permit permissionless entry under pseudonymity. Second, cooperative ownership is purchased; well-designed protocol ownership is earned through contribution (staking, hardware deployment, service delivery), a property Scholz (2016) identifies as the platform-cooperativism premise. The governance concentration data presented in Section 3 indicate that most protocols have not collapsed the capital-labor boundary in practice: holding HHI values span from cooperative-grade distribution (Anyone Protocol 0.040) to single-party control. Third, cooperatives enforce through bylaws, elected boards, and courts; protocols enforce through code, gaining credible neutrality at the cost of contextual judgment unless appeal mechanisms are designed in. Fourth, cooperatives operate within one jurisdiction's cooperative law; protocols operate across jurisdictions simultaneously, making Ostromian polycentricity (developed in Section 2.6) a structural necessity rather than a design preference. Fifth, cooperative fiscal discipline operates through dues and patronage dividends; protocol fiscal discipline operates through emissions and burns, with the Spend-to-Reward ratio the protocol equivalent of cooperative fiscal sustainability.

These distinctions define the contribution space: protocols inherit the cooperative's legitimacy challenge (governing collective production without a centralized principal) while adding permissionless entry, programmable enforcement, and borderless jurisdiction as additional design constraints. The polycentric framework in Section 2.6 takes up this expanded design space; the empirical results in Section 3 indicate how far current protocols remain from satisfying it.

2.5 DePIN and Tokenized Physical Infrastructure: Demand Coupling and Service Verification

DePIN expand tokenized coordination into the physical world (Bane & Gala, 2025, 2026): hardware deployment, service provision, and maintenance. DePIN intensifies the institutional nature of tokenomics because it coordinates capital formation (hardware investment), labor (operations), and service quality (SLA performance). The result is a system that resembles public infrastructure governance more than application monetization.

Recent taxonomies formalize DePIN as a distinct coordination category with three architectural layers spanning distributed ledger, cryptoeconomic, and physical infrastructure dimensions (Ballandies et al., 2023), while Alshater (2026) synthesizes the token flywheel mechanism through which burn-and-mint equilibria bootstrap supply-side hardware deployment.

A key design pattern in DePIN is dual-asset economics (Voshmgir, 2020) separating speculative governance/stake tokens from stable-value usage credits. Helium provides a prominent example: network usage is paid in Data Credits, a non-transferable utility token used for service payments and onboarding actions.
This dual-asset structure motivates a general principle developed later: stable-value usage pricing reduces user friction, while governance/stake tokens enable long-term alignment and security. Separate simulation work stress-tests DePIN mechanism designs under adversarial conditions using cadCAD (a Python-based simulation framework), demonstrating that mechanism-level choices (emission

model, slashing parameters) produce macro-level institutional outcomes that the normative framework developed here can evaluate.

2.6 Constitutional Political Economy and Polycentric Governance

2.6.1 Constitutions as Rule-Selection Under Constraints

Constitutional political economy (Buchanan & Tullock, 1962) treats political rules as objects of analysis and choice rather than fixed background conditions. Buchanan and Tullock frame constitutions as foundational constraints shaping later policy choices and collective decisions; this resonates with tokenized systems where "hard parameters" (e.g., inflation bounds, timelocks, emergency powers) are embedded in code. Buchanan and Tullock explicitly describe their work as analyzing the political organization of a society of free persons, using economic reasoning to examine constitutional constraints.

This literature (Buchanan & Tullock, 1962) informs later chapters' distinction between a protocol's "constitutional layer" (hard-to-change constraints) and "policy layer" (adjustable parameters).

2.6.2 Ostrom and Polycentricity

Ostrom's (1990) work provides one of the strongest empirical foundations for governance of commons and public service industries. Her concept of polycentric governance emphasizes multiple overlapping decision centers, local experimentation, and rule diversity. Ostrom's AER article explicitly characterizes polycentric systems as alternatives to the binary "market vs state" framing, offering an analytical framework for complex economic systems with multi-level governance. Van Vulpen and Jansen (2023) map Ostrom's eight design principles to DAO governance structures, providing a standardized evaluation framework for collective action sustainability; the scoring rubric developed here (Table 1) extends their mapping by adding four non-Ostromian traditions and operationalizing the resulting criteria through quantitative governance concentration data. Li and Chen (2024) conceptualize DAOs as collective stewards of digital commons, adapting Ostrom's design principles to emphasize equitable distribution of decision-making authority; the governance concentration data in Section 3 test whether operating protocols achieve the distributional outcomes this framing requires. De Filippi et al. (2024) analyze overlapping decision-making structures in blockchain systems through Ostrom's polycentricity framework, identifying structure, process, and outcome dimensions that map onto the governance concentration, delegation, and participation metrics measured here.

Polycentricity (Ostrom, 1990) is particularly salient for DePIN, where service conditions differ across geography and where local knowledge is crucial for operational decisions. The paper later proposes polycentric sub-DAOs/councils as governance primitives for range-based or region-based tuning.

2.7 Political Philosophy as Legitimacy Specification

Five normative traditions constrain how legitimate institutional design should look. Kant's (1785/1997) publicity principle holds that political maxims must be capable of public avowal and that coercion (slashing, bans, parameter changes) is legitimate only when grounded in transparent and non-arbitrary rules. Republican freedom as non-domination (Pettit, 1997; Stanford Encyclopedia of Philosophy, 2025) reframes legitimacy as protection against arbitrary power rather than mere non-interference, translating into design as contestability through appeals, oversight, and bounded enforcement discretion. The capability approach (Sen, 1999; Nussbaum, 2011) evaluates social arrangements not by aggregate output but by the substantive opportunities individuals have to achieve valued doings and beings, a frame Yanaka and Heeks (2023) extend to digital platform labor and which applies directly to DePIN operators whose hardware lock-in constrains exit options. Hayek's (1945) "Use of Knowledge in Society" argues that dispersed local information cannot be replicated by centralized planning, providing both a critique of static parameter-setting and a positive case for governance designs that incorporate edge-level signals.

A fifth tradition addresses the data-and-identity infrastructure that tokenized networks require for sybil resistance, proof-of-service, and reputation. Nissenbaum's (2010) framing of privacy as contextual integrity treats appropriate information flow as governed by context-specific norms; Floridi (2013) extends this into a broader information ethics. Together, these frameworks underwrite the design claim that token systems relying on ever-expanding telemetry risk becoming disciplinary infrastructures unless constrained by contextual integrity and proportionality. The five traditions are not isolated lenses but complementary specifications of legitimacy, jointly defining the design constraints the framework operationalizes in Section 2.9 and the scoring rubric in Table 1.

2.8 Synthesis and Research Gap

Across economics, computer science, political philosophy, and institutional analysis, a shared conclusion emerges: tokenized systems create constitutions and policies in code (Werner et al., 2022). Yet existing literature rarely provides a unified framework linking (i) philosophical legitimacy constraints, (ii) tokenomics mechanism design, and (iii) measurable outcomes in sustainability, governance quality, resilience, and social welfare.

The contribution is threefold:

an exclusion methodology (Section 3.2) that identifies 69 addresses controlled by protocols themselves but appearing on holder lists, correcting a measurement artifact in prior work,

a cross-sector analysis (Section 3) documenting that sector membership predicts concentration after adjusting for protocol age, valuation, and insider allocation, while initial allocation, maturity, and subsidy intensity do not, and

a normative framework (Sections 2.9–2.10) translating five political-philosophical traditions into observable design criteria through a scoring rubric that other researchers can apply.

2.9 Theoretical Framework

2.9.1 Tokenomics as Applied Political Philosophy

Tokenomics is a political-constitutional problem, implicit in protocol debates but rarely formalized. Token systems allocate:

rights (access, governance influence, reward eligibility),

duties (staking requirements, performance obligations, compliance burdens), resources (rewards, treasury allocations), and sanctions (slashing conditions, enforcement procedures, dispute resolution).

The philosophical question - “What makes a rule system legitimate?” - therefore becomes an engineering question: “What mechanism properties produce legitimacy under adversarial conditions and pluralistic values?”

To integrate cross-disciplinary perspectives, the framework distinguishes three levels:

Normative constraints (philosophical commitments): publicity, non-domination, floor protection, capability enhancement, contextual integrity.

Institutional mechanisms (tokenomics primitives): supply, distribution, sinks, incentive design, governance architecture, enforcement and appeals.

Outcomes: economic sustainability, resilience, political legitimacy (participation, contestability, concentration), and socially meaningful gains.

2.9.2 Formal Definitions and Notation

Definition 1 (Sink and Faucet)

A faucet is total net token issuance distributed as rewards or other liquid supply increases over a given time window. A sink is any mechanism that permanently or temporarily removes tokens from circulation: burns, buybacks that retire tokens, or locks that reduce tradeable supply. A system exhibits "sink-before-faucet" discipline when sinks are deployed and observable before (or alongside) large-scale faucets, providing the demand-coupling discipline that Hypothesis 3 formalizes.

Definition 2 (Spend-to-Reward Ratio, S2R)

The Spend-to-Reward Ratio (S2R) is the ratio of demand-side spending (tokens removed via sinks) to supply-side issuance (tokens distributed via faucets) over a given time window. S2R serves as a fiscal sustainability indicator: values below 1.0 mean issuance exceeds spending and the system is subsidy-dependent; values above 1.0 mean spending offsets issuance and the system is demand-financed. Higher S2R implies tighter coupling between usage and token scarcity.

Definition 3 (Outcome Domains)

Outcomes are grouped into three domains. Economic outcomes include sustainability (revenue and fee flow), market quality (liquidity depth and slippage), resilience (drawdowns and recovery time), and efficiency. Political outcomes include participation rates, concentration (measured by HHI and Gini), procedural justice (appeals, timelocks, contestability), and rule stability. Social outcomes include capability gains, geographic and socioeconomic inclusion, safety, and privacy-preserving participation. The Section 3 empirics use political-domain metrics; economic and social metrics are operationalized in companion studies.

2.9.3 Conceptual Model: From Philosophy to Mechanisms to Outcomes

The conceptual model of tokenomics as applied political philosophy organizes the framework across four layers, shown in Figure 1.

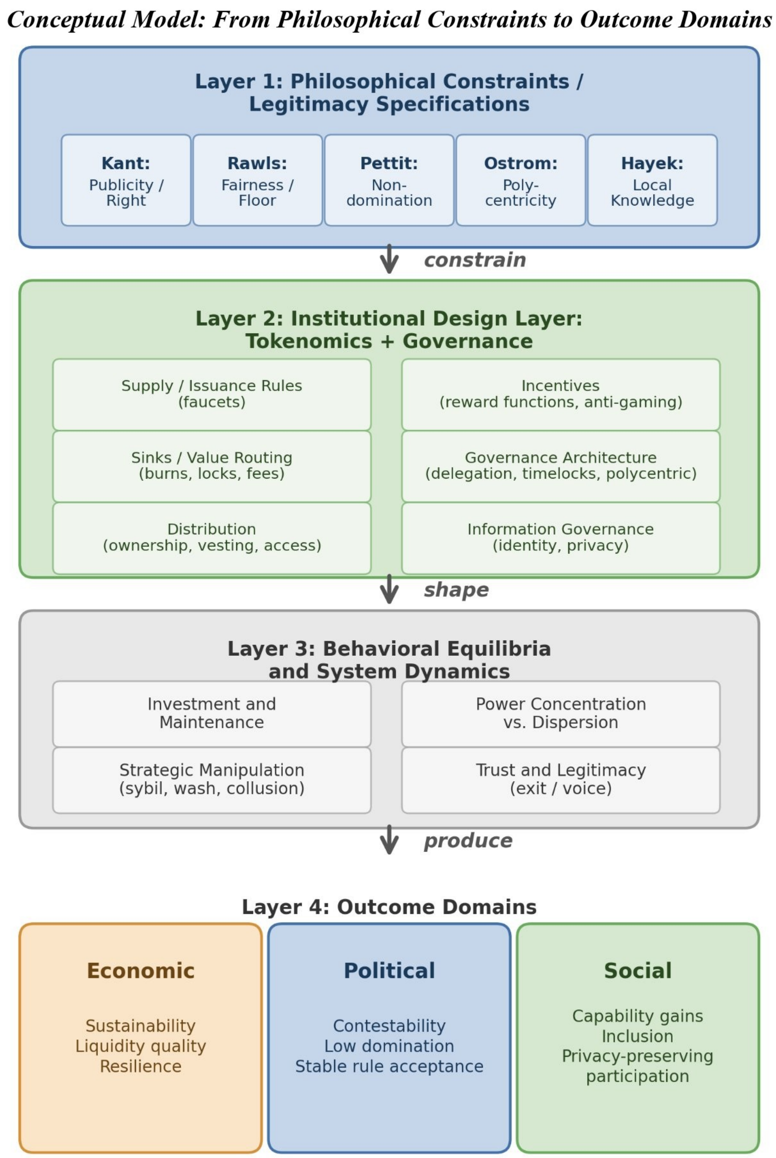


Figure 1. *Conceptual model: from philosophical constraints through institutional design to outcomes. Philosophical traditions (Layer 1) constrain institutional design choices (Layer 2), which shape behavioral equilibria (Layer 3) and produce measurable outcomes across economic, political, and social domains (Layer 4).* Reading the figure: the arrows trace the design pipeline from philosophical commitments through measurable outcomes; each lens in the rubric is a constraint on permissible designs at the corresponding stage.

The model's purpose is explanatory and testable: it identifies where philosophical principles enter as constraints, how they shape design choices, and how those design choices influence outcomes through strategic behavior and system dynamics.

2.9.4 Design Hypotheses (Normative, Testable in Principle)

The theoretical framework yields five core design hypotheses linking political-philosophical legitimacy constraints to measurable institutional properties. This paper evaluates these hypotheses primarily using a governance concentration cross-section; hypotheses requiring operational demand, incentive dynamics, or participant outcome data are evaluated in future work or specified as future measurement priorities.

Hypothesis 1 (Publicity - Legitimacy Link)

Systems that satisfy a strong publicity constraint: transparent rules, published rationales, and procedurally explicit enforcement, exhibit fewer legitimacy crises and higher participation persistence than systems with opaque or discretionary policy.
Justification: Kant's publicity principle suggests that political maxims requiring secrecy to succeed are illegitimate; in token systems, secrecy increases perceived arbitrariness and reduces willingness to invest in the future under rule uncertainty.

Hypothesis 2 (Contestability - Non-Domination)

Introducing contestability mechanisms: appeals with bounded resolution time, independent oversight, and publishable enforcement criteria, reduces domination risk and improves long-run operator/user retention, conditional on adequate measurement integrity.
Justification: Republican freedom as non-domination requires protection against arbitrary interference; appeals and oversight operationalize that requirement by constraining enforcement discretion.

Hypothesis 3 (Sinks Before Faucets)

Holding other factors constant, systems that deploy measurable sinks (burns/locks/fee routing) before large-scale emissions have higher fiscal sustainability (S2R) and reduced boom-bust amplitude than systems that rely primarily on emissions to bootstrap growth.
Justification: In repeated interaction settings, participants discount future rewards when dilution is unconstrained; sinks create credible coupling between real usage and token scarcity/treasury strength.

Hypothesis 4 (Service over Presence in DePIN)

In DePIN settings, shifting reward weight from "presence/coverage" to "SLA-verified service delivered" reduces oversupply and increases demand-weighted efficiency and resilience, relative to presence-mining regimes.

Justification: Presence-only incentives reward capital deployment irrespective of marginal social value; service incentives approximate payment for delivered public service capacity, aligning rewards with demand.

Hypothesis 5 (Polycentric Adaptation)

Polycentric governance structures (multiple centers with localized authority and budgets) improve adaptation to heterogeneous service environments (geography, range demand, local constraints) and reduce governance deadlock compared to monocentric governance.
Justification: Ostrom's polycentric governance emphasizes multiple overlapping decision centers as an efficient response to complex systems.

2.9.5 Implications for Cross-Program Relevance

Political theory programs can treat tokenomics as a novel constitutional domain: public rules with automatic enforcement and low-cost exit.

Economics/public policy programs can interpret tokenomics as fiscal and regulatory policy under open entry and adversarial manipulation, requiring mechanism design and institutional economics.

Computer science/security programs can frame tokenomics as incentive-aligned security and reliability engineering where verification systems ($\mathscr{M}$) are first-class determinants of institutional feasibility.

STS and law can analyze how "code as governance" transforms legitimacy, accountability, and liability structures, and how transparency and contestability affect social acceptance.

Lens	0 (Absent)	1 (Minimal)	2 (Partial)	3 (Exemplary)
Kantian Publicity	No public governance docs	Rules exist but hard to find	Public rules + rationale	Public rules + deliberation + justification
Rawlsian Fairness	No floor-raising mechanism	Basic airdrop only	Graduated fee tiers or grants	Systematic equity mechanisms + measured gaps
Pettit Contestation	No appeals or checks	Informal complaints possible	Formal appeal mechanism exists	Independent arbitration + veto rights
Ostrom Polycentricity	Single decision center	Advisory subcommittees	Functional subDAOs with budgets	Nested governance with local autonomy
Hayek Knowledge	Fixed parameters	Admin-adjusted parameters	Automated price signals	Demand-coupled mechanisms + edge feedback

Table 1. *Scoring rubric: philosophical lens evaluation (0 = absent, 3 = exemplary). Traditions glossed: Kantian Publicity (transparent rule justification), Rawlsian Fairness (protection of the least advantaged), Pettit Contestation (freedom from arbitrary power), Ostrom Polycentricity (distributed decision-making), Hayek Knowledge (channeling local information).*

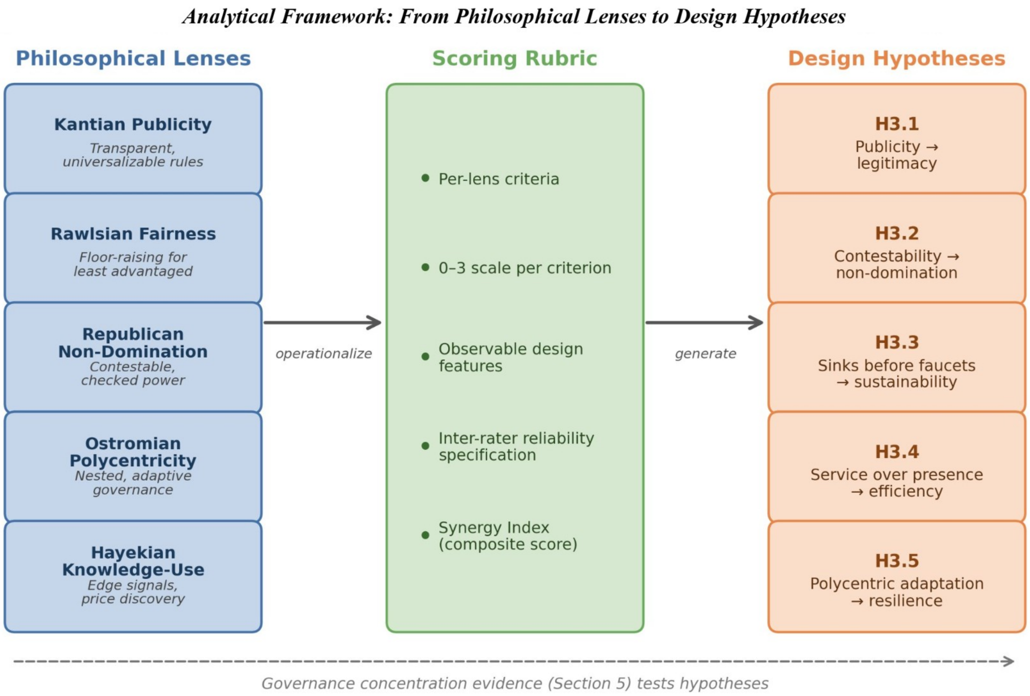


Figure 2. *Normative framework architecture: five philosophical lenses are operationalized through a scoring rubric into five testable design hypotheses. Dashed arrow indicates the empirical feedback loop from governance concentration evidence (Section 3) and case evidence developed separately.* Reading the figure: each axis represents a separate philosophical lens; protocols satisfying multiple lenses occupy upper-right regions.

2.10 Framework Operationalization

2.10.1 Approach

The framework is operationalized through three components: (1) a structured dataset of protocol and DAO design variables; (2) a philosophy-to-design scoring rubric translating the five normative lenses into observable criteria; and (3) a governance concentration cross-section measuring token distribution across 40 protocols. This paper develops the framework and rubric, and presents the governance concentration cross-section as an empirical illustration. The full quantitative pipeline (regression, panel, and event-study analyses) and qualitative case evidence are specified for future work and developed in separate studies.

2.10.2 Units of Analysis and Sampling Strategy

Holder-list cutoff: the top-1,000 holder cutoff follows established convention in the token-governance concentration literature (Sai et al., 2021 use comparable cutoffs across blockchain governance studies). The cutoff captures the holder population materially relevant to governance decisions; long-tail holders below this threshold collectively hold less than five percent of supply across all sample protocols. Robustness across cutoffs is supported by the Top-1%, Top-5%, and Top-10% columns in Table 4: rank-ordering across protocols is preserved at all three cutoffs (Spearman rho > 0.95 between any two columns), indicating that absolute HHI values may shift modestly with deeper-tail inclusion but cross-protocol comparison remains robust to the cutoff choice. For protocols with fewer than 1,000 unique non-zero holders after exclusion (e.g., ZRO with N = 989), the actual holder count is reported; the 1,000-holder cutoff serves as a ceiling rather than a floor.

Primary unit: **protocol-token system** (including its governance institutions). Secondary unit: **governance episodes/events** (emission changes, fee switches, unlocks, enforcement controversies, governance reforms).

Protocols were selected to maximize variation across three dimensions: sector (15 DeFi, 15 DePIN, 8 L1/L2 infrastructure,2 failed or stagnated protocols included to avoid studying only survivors), maturity (2017-2024 launches), and governance model (token-weighted, delegate, futarchy). The expanded 40-protocol sample was constructed by adding protocols from the Dune Analytics ecosystem that met three criteria: (i) publicly traded governance token, (ii) holder data queryable via Dune or Helius DAS API, and (iii) at least 50 non-zero-balance holders after protocol-controlled address exclusion. MetaDAO is included as the sole futarchy-governance protocol in the sample. Solana-native tokens (META, DRIFT, GRASS, W) required a separate data collection pipeline (Helius DAS API) because the primary indexer undercounted holders by two orders of magnitude; details are in Supplementary File S6.

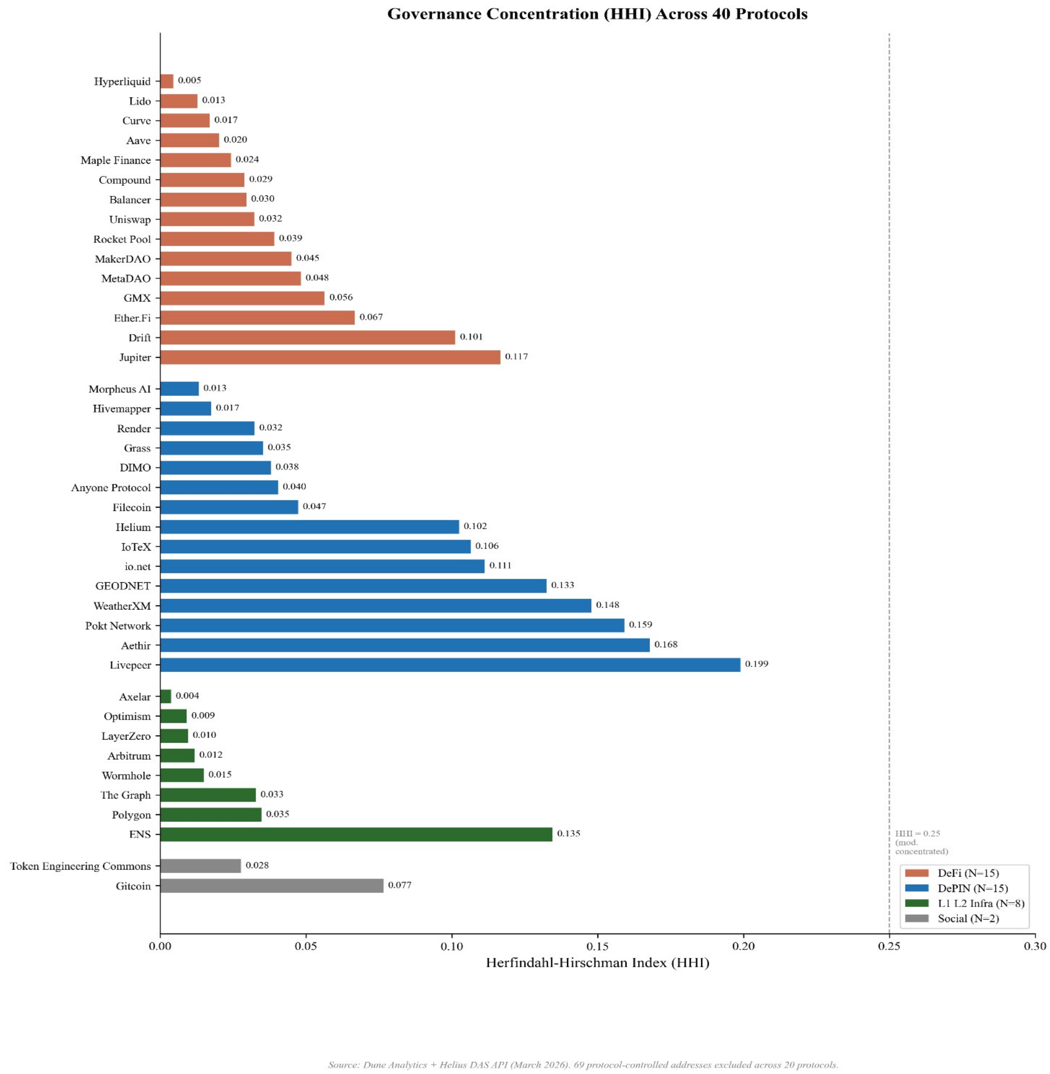


Figure 3. *Holding-based Herfindahl-Hirschman Index (HHI) across 40 governance tokens (March 2026 snapshot, top-1,000 holders, protocol-controlled addresses excluded). Dashed lines indicate category means. Protocols sorted by category then HHI ascending.* Reading the figure: bars show pre-exclusion HHI; the post-exclusion overlay indicates how much measured concentration was protocol-controlled rather than governance-relevant.

Inequality (Gini) is near-universal across all sectors (range 0.45 to 0.99), while concentration (HHI) varies by an order of magnitude (0.004 to 0.199 after excluding protocol-controlled addresses). Token wealth inequality is a structural feature of token systems, but governance concentration, how far power concentrates among few addresses, is design-sensitive. Governance structure is a constitutional choice, not an afterthought. Prior work applying the Gini coefficient to eight major cryptocurrencies found inequality levels comparable to real-world economies (Sai, Buckley, & Le Gear, 2021); because permissionless address creation produces millions of near-zero-balance wallets that inflate Gini toward 1.0 regardless of governance structure, the near-universal inequality this sample documents (Gini 0.52 to 0.99) is a measurement artifact of ledger architecture rather than a governance signal, reinforcing the case for HHI as the primary concentration metric.

The Anyone Protocol (HHI 0.040; Table 4) demonstrates that governance concentration is a design choice, not an inevitability of DePIN architecture. Five DePIN protocols (Render 0.032, Grass 0.035, DIMO 0.038, Anyone 0.040, Filecoin 0.047) achieve concentration levels at or below the DeFi median, indicating that low concentration is achievable in hardware-coordination protocols.

Sampling principles: include multiple categories (L1/L2, DeFi, infra/oracles, DePIN, social)

include "dead" or stagnated projects to reduce survivorship bias

stratify by chain and epoch to control for regime effects

2.10.3 Data Sources (Replicable, Multi-Disciplinary)

Four data streams inform the analysis: on-chain data (Dune Analytics, Helius DAS API), protocol documentation (whitepapers, governance forums, parameter docs), analytic aggregators (Token Terminal, DeFiLlama, CoinGecko), and public academic and policy sources. A systematic data coverage gap between DeFi and DePIN protocols (Token Terminal covers approximately 100% of DeFi but only 25% of DePIN) constrains cross-sector regression analysis. Full source provenance, exclusion methodology, the protocols Token Terminal does not cover, and the resulting partial-data subsets are documented in Supplementary File S6.

2.10.4 Codebook and Rubric

Insider taxonomy: the analysis distinguishes two related but distinct constructs. Insider allocation (insider_pct) is the regression covariate measured at token generation event (TGE), defined as team_pct + investor_pct from primary tokenomics sources. Insider count fraction in top-10 holders is a descriptive mechanism variable measured post-distribution from holder snapshots, defined as the post-exclusion fraction of wallets in the top-10 classified as insider via a three-tier procedure: Tier 1 uses Blockscout verified-contract labels and Dune exchange tags; Tier 2 uses contract bytecode matching for SafeProxy, GnosisSafe, and vesting contracts; Tier 3 uses Etherscan named-address matching against documented foundation, treasury, and team wallets. Protocol-controlled addresses (PCAs) are excluded from holding HHI computation entirely at the top of the pipeline (69 addresses across 20 protocols documented in exclusions_log.csv) and are not counted as insiders for the purposes of the count fraction. Solana-native protocols carry an acknowledged labeling gap: io.net classifies as zero of ten insiders because

Solana lacks Blockscout-style contract labels at the scale available on Ethereum and EVM L2s. This is a classification limitation rather than evidence of distributed governance and is flagged in the section 4.8 limitations.

We code each protocol across standardized dimensions via a machine-readable table capturing metadata, supply and issuance rules, distribution, value routing and sinks, staking and lockup parameters, governance architecture, decentralization metrics, incentive windows, and data/identity posture, plus computed philosophy lens scores (0–3) and evidence links. Full field definitions and the codebook schema are documented in Supplementary File S1.

2.10.5 Philosophy Scoring Rubric: Translating Normative Theory into Observable Criteria

The scoring rubric is a second machine-readable table (Supplementary File S2) capturing lens_id, criterion_id, criterion definitions, a 0–3 scoring scale per criterion, and default weights; a third table (Supplementary File S3) stores protocol × criterion scores with evidence links, scorer initials, and dates. Each protocol is scored by gathering evidence (documentation, forum threads, code references), assigning 0–3 to each criterion, computing each lens score as a weighted mean, and aggregating into a Synergy Index across lenses with pre-specified weights.

Reliability is established by double-coding at least 10% of protocols to estimate inter-rater agreement (Cohen's κ), with adjudication rules where κ is low; construct validity is established by comparing rubric scores to independent indicators including governance concentration, frequency of rule changes, and appeals outcomes. The operationalization is designed to be legible across disciplines, mapping political-philosophical criteria to measurable institutional features (appeals, publicity, concentration) and enabling empirical tests.

Table 2 demonstrates the rubric applied to three protocols spanning the DeFi–DePIN divide; full scoring sheets for all 12 scored protocols, with evidence links, are provided in Supplementary File S3. Scores are assigned by a single coder; inter-rater reliability testing with a second scorer is specified as future work.

Protocol	Publicity	Fairness	Non-Dom.	Polycentric	Knowledge	Key evidence note
Uniswap (DeFi)	2	1	2	1	1	Forum + Snapshot + on-chain vote; HHI 0.114; delegate concentration high
Hyperliquid (DeFi)	2	1	1	0	2	Zero VC + 31% airdrop; HHI 0.005; Assistance Fund auto-burns 97% of fees

Protocol	Publicity	Fairness	Non-Dom.	Polycentric	Knowledge	Key evidence note
Helium (DePIN)	2	1	2	3	3	Helium Improvement Proposal (HIP) process public; subDAO structure; HHI 0.102; HIP-147 reform
GEODNET (DePIN)	2	1	1	1	2	B2B subscription-burn (Real-Time Kinematic GPS service); HHI 0.132; Snapshot DAO; demand-coupled
Anyone (DePIN)	1	1	1	1	1	Low HHI (0.040); early-stage governance; limited formal process

Table 2. *Illustrative rubric scores (0 = absent, 1 = minimal, 2 = partial, 3 = exemplary). Scores are by a single coder; full evidence sheets in* Supplementary File S3*. Columns: Publicity (Kantian; transparent rules with published rationales), Fairness (Rawlsian; floor protection in distribution), Non-Dom. (Pettit; formal contestability infrastructure protecting from arbitrary interference, scored independently of holding concentration), Polycentric (Ostrom; multiple overlapping decision centers), Knowledge (Hayek; reliance on dispersed local knowledge in operations).*

2.10.6 Outcome Variables

Three primary metrics are used. The Herfindahl-Hirschman Index (HHI), ranging from 0 (dispersed) to 1 (monopoly), measures overall token concentration; values above 0.25 indicate high concentration by antitrust standards. Section 3.3 notes no protocol exceeds this threshold. The Gini coefficient captures distributional inequality. Top-N ratios (Top-1, Top-5, Top-10 share) complement HHI by identifying specific whale addresses.

The Synergy Index (mean of five lens scores: Kantian publicity, Rawlsian fairness, republican non-domination, Ostromian polycentricity, Hayekian knowledge-use; each 0–3 per Table 1) measures institutional design intent across normative traditions. Three dimensions from the literature review (capabilities, contextual integrity, cosmopolitan inclusion) are reserved for future evaluation due to non-comparable cross-protocol measurement.

Full metric definitions, computation procedures, and Dune query specifications are provided in Supplementary File S1.

2.10.7 Testability and Empirical Pipeline

The five design hypotheses are testable in principle through cross-sectional governance analysis (this paper), longitudinal panel models, event studies around governance decisions, and qualitative process tracing. This paper reports the cross-sectional governance concentration analysis (Section 3). The full quantitative pipeline, including panel regressions, event-study specifications, and robustness protocols, is documented in supplementary material (Supplementary File S5) for replication and extension by future researchers.

2.10.8 Ethical and Practical Considerations

All data are drawn from public sources: on-chain transactions, published governance proposals, and open protocol documentation. No private or proprietary data are used. The scoring rubric relies on publicly observable design features. We compute governance token distribution data from Dune Analytics multi-chain queries. The author is employed by Borderless Capital, a cryptocurrency-focused investment firm, and may hold or have held positions in some protocols analyzed; a complete disclosure is provided in the Conflict of Interest statement accompanying this submission.

3 Results

3.1 Scope and Claims Boundary

The cross-section covers 40 protocols across four sectors (DePIN, DeFi, infrastructure, social). Table 3 delimits the claims boundary: all findings are descriptive within the observed scope. All claims are descriptive within the observed scope. The governance data represent a March 2026 snapshot, not a time series. Causal claims about the relationship between institutional design and governance outcomes require the longitudinal evidence beyond the scope of this paper. All governance token distributions were queried from Dune Analytics multi-chain queries. Concentration metrics are point-in-time snapshots; trajectory analysis requires the longitudinal extension specified in the Future Research discussion (Section 2.10).

Claim	Evidence Grade	Pointer
Governance concentration is descriptively higher in DePIN than DeFi	Descriptive	Table 4
The scoring rubric translates philosophy into measurement	Methodological	Table 1 + Supplementary File S1
Five core design hypotheses follow from the normative framework	Theoretical	Section 4.1

Table 3. *Claims boundary: what this paper does and does not claim.*

3.2 Cross-Protocol Concentration

A measurement caveat: the 0.25 HHI antitrust threshold is borrowed from market-concentration analysis (DOJ Horizontal Merger Guidelines) and is reported here as a benchmark for descriptive interpretation only. A market-concentration threshold is not equivalent to a democratic-legitimacy threshold; a protocol may have HHI below 0.25 while remaining politically dominated through quorum rules, delegation patterns, or proposal-rights concentration. The HHI captures one dimension of concentration (post-exclusion holding distribution) and is not a sufficient condition for distributed governance.

Governance concentration spans two orders of magnitude across the 40-protocol sample (Table 4): HHI ranges from 0.005 (Hyperliquid) to 0.199 (Livepeer), with a median near 0.04. DePIN protocols cluster in the upper half of this distribution. Livepeer is the clearest structural outlier; excluding it leaves a distribution in which DePIN still exhibits higher concentration than DeFi but without a dominant leverage point.

Concentration metrics are computed from Dune Analytics multi-chain queries and the Helius DAS API for Solana SPL tokens (March 2026 snapshot, top-1000 holders per protocol), after excluding 69 protocol-controlled addresses across 20 protocols (staking contracts, exchange custodians, vesting locks, protocol treasuries; see Supplementary File S6). Solana-native tokens use getTokenAccounts with page-based pagination to correct systematic undercounting in Dune's indexed tables. Replication scripts and input CSVs are listed in Supplementary File S5.

Three additional DePIN protocols were added to improve statistical power for the subsidy analysis: Aethir (ATH, Ethereum, GPU compute, $156M annualized revenue (Token Terminal, April 2026)), Hivemapper (HONEY, Solana, mapping), and io.net (IO, Solana, GPU compute, $21M revenue). Governance concentration was computed using the same methodology (Dune for ERC-20 tokens, Helius for Solana SPL tokens, top-1000 holders, exchange-labeled and protocol-controlled addresses excluded).

Protocol	Token	Category	HHI	Gini	Top-1%	Top-10%	N
Morpheus AI	MOR	DePIN	0.0310	0.88	11.4%	43.3%	994
Render	RENDER	DePIN	0.0323	0.89	10.4%	47.0%	1000
Grass	GRASS	DePIN	0.0352	0.76	16.0%	38.2%	1000
DIMO	DIMO	DePIN	0.0379	0.87	13.9%	45.0%	999
Anyone Protocol	ANYONE	DePIN	0.0404	0.79	13.3%	47.5%	1000
Filecoin	FIL	DePIN	0.0473	0.79	18.0%	45.6%	1000
Helium	HNT	DePIN	0.1024	0.98	24.5%	79.4%	1000
IoTeX	IOTX	DePIN	0.1065	0.99	19.8%	90.6%	1000
GEODNET	GEOD	DePIN	0.1325	0.96	24.9%	81.8%	1000
WeatherXM	WXM	DePIN	0.1479	0.89	35.0%	77.9%	999
Pokt Network	POKT	DePIN	0.1592	0.96	34.8%	77.2%	1000
Livepeer	LPT	DePIN	0.1989	0.94	42.9%	73.7%	998
Hyperliquid	HYPE	DeFi	0.0045	0.73	2.2%	14.5%	1886
Lido	LDO	DeFi	0.013	0.52	7.9%	27.7%	189
Maple Finance	MPL_SYRUP	DeFi	0.0242	0.88	8.5%	40.6%	998
Compound	COMP	DeFi	0.0275	0.86	14.1%	32.4%	1000
Balancer	BAL	DeFi	0.0295	0.88	12.4%	38.0%	990
Rocket Pool	RPL	DeFi	0.0391	0.87	11.5%	52.7%	999
MakerDAO	MKR	DeFi	0.0450	0.90	11.4%	56.2%	1000
MetaDAO †	META	DeFi	0.0482	0.90	19.6%	41.4%	1000
GMX	GMX	DeFi	0.0564	0.93	17.5%	60.2%	999
Aave	AAVE	DeFi	0.020	0.88	7.5%¶	~~47.6%~~3	1000

Protocol	Token	Category	HHI	Gini	Top-1%	Top-10%	N
						4.9%¶	
Ether.Fi	ETHFI	DeFi	0.0668	0.95	19.0%	59.9%	1000
Uniswap	UNI	DeFi	~~0.032~~0.010¶	0.89	4.2%¶	~~52.1%~~22.9%¶	1000
Drift	DRIFT	DeFi	0.1011	0.94	27.0%	61.7%	1000
Jupiter	JUP	DeFi	0.1166	0.98	25.1%	75.0%	1000
Curve	CRV	DeFi	~~0.1706~~ 0.017¶	0.89	6.7%	60.7%	1000
Axelar	AXL	L1/L2/ Infra	0.0277	0.85	9.2%	42.7%	994
LayerZero	ZRO	L1/L2/ Infra	0.0147	0.89	4.1%	30.5%	989
Wormhole	W	L1/L2/ Infra	0.0149	0.87	7.0%	23.8%	1000
Polygon	POL	L1/L2/ Infra	0.0348	0.92	12.1%	49.5%	998
Arbitrum	ARB	L1/L2/ Infra	0.012	0.83	8.6%¶	~~47.2%~~22.4%¶	1000
The Graph	GRT	L1/L2/ Infra	0.033	0.92	13.1%¶	~~59.7%~~40.2%¶	1000
Optimism	OP	L1/L2/ Infra	0.009	0.89	12.1%¶	~~60.8%~~29.6%¶	1000
ENS	ENS	L1/L2/ Infra	0.1345	0.94	26.7%	75.5%	999
Token Engineering Commons ‡	TEC	Social	0.0278	0.91	9.2%	44.2%	742
Gitcoin ‡	GTC	Social	0.0766	0.94	17.6%	64.3%	1000

Table 4. *Governance concentration cross-section (March 2026 snapshot, Dune Analytics + Helius DAS API, top-1,000 holders per token, 69 protocol-controlled addresses excluded across 20 protocols). Sorted by category then HHI ascending. Balancer and Curve use vote-escrowed governance mechanisms (veBAL, veCRV). Reported HHI values measure raw token holding concentration, not voting power. Vote-escrowed concentration is expected to be higher due to lock-duration multipliers, analogous to the stkAAVE caveat noted above. Hyperliquid is included in Table 4 but excluded from the subsidy regression because the protocol has no token emissions; 97% of trading fees are routed to an Assistance Fund that purchases and permanently burns HYPE tokens, treated as a sink mechanism in the framework. Three additional DePIN protocols (Aethir, Hivemapper, io.net) appear in the cross-sector statistical tests reported in Section 3.3 with HHI values computed (0.171, 0.020, and 0.111 respectively) but lack the holder-level Gini and percentile data required for Table 4 reporting; their HHI values are noted in the relevant analysis sections.*

Table 4 footnotes:

† = futarchy governance (MetaDAO).

‡ (in protocol name column) = survivorship-bias control (TEC, Gitcoin).

¶ (added in R1 revision) = updated post-exclusion value: HHI for Curve reflects raw CRV holdings (veCRV-weighted concentration is approximately 0.171 due to lock-duration multipliers); HHI for Uniswap reflects post-burn-rule exclusion of 0x000...dead (102.46M UNI, 11.27% of supply); Top-1% and Top-10% columns report post-exclusion shares for marked rows (AAVE, UNI, ARB, GRT, OP) per R1 reviewer methodological consistency request.

HHI ranges from 0.005 to 0.199 across the full sample. Among DeFi governance tokens, HHI ranges from 0.005 (Hyperliquid, after excluding the Assistance Fund and burn addresses) to 0.171 (Curve). DePIN protocols span a similar range: 0.017 (Hivemapper) to 0.199 (Livepeer); Morpheus AI registers 0.031 after excluding four protocol-controlled addresses.

Several protocols previously appearing highly concentrated show moderate concentration once measurement artifacts are removed. After correcting for holder-list truncation (Solana tokens) and protocol-controlled address contamination (19 protocols), the DePIN-DeFi concentration gap narrows substantially: DRIFT (0.419 uncorrected to 0.101 corrected, reflecting Helius enumeration of 27,888 holders versus 104 from Dune), W (0.126 to 0.015), and GRASS (0.132 to 0.035). The exclusion methodology (Supplementary File S6) is particularly consequential for protocols with large foundation treasuries or exchange custodian addresses in the top-10 holders.

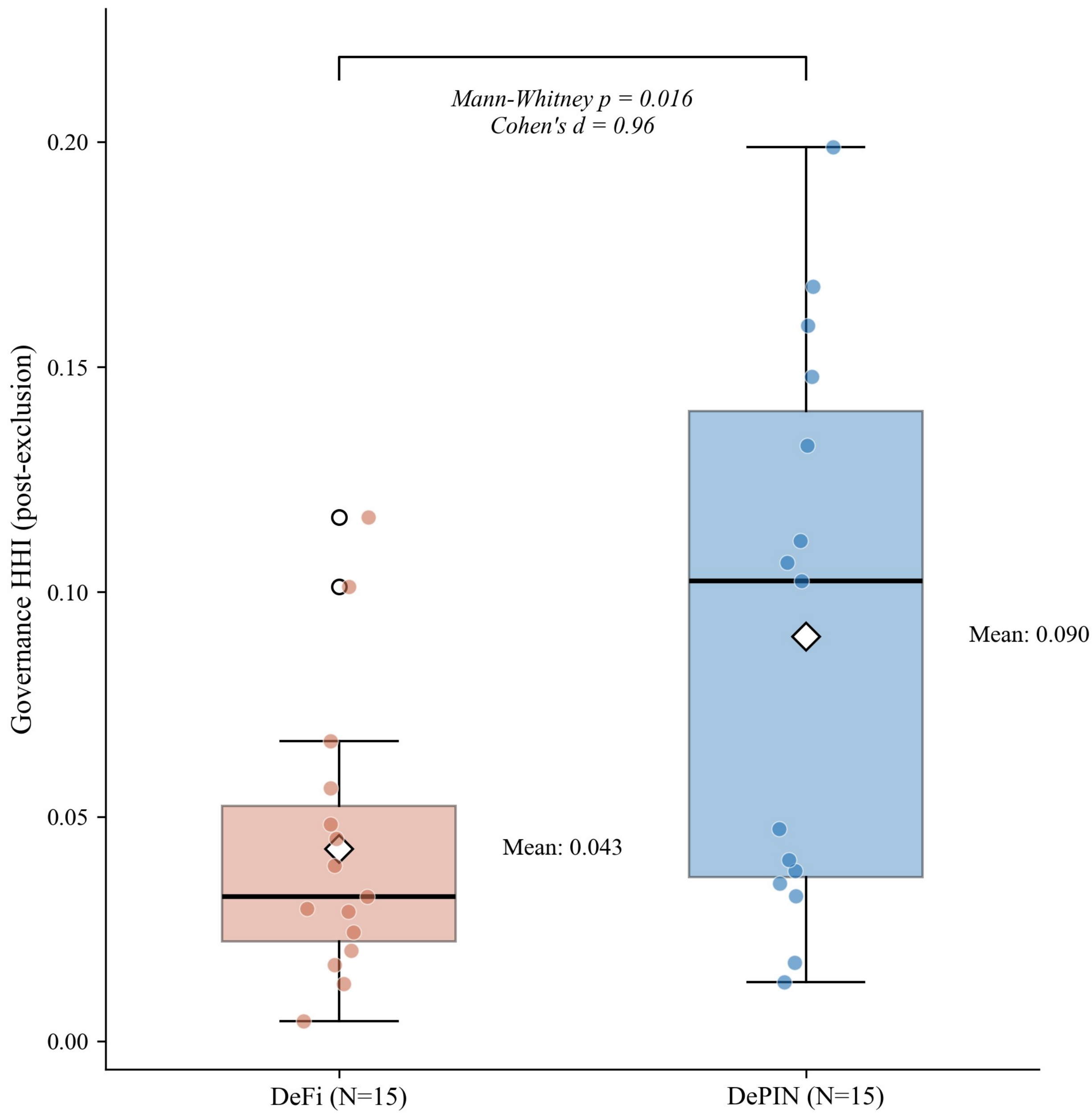
Several protocols demonstrate that low governance concentration is achievable across sectors. Hyperliquid (HHI 0.005), Uniswap (0.010), and Lido (0.013) in DeFi; Optimism (0.009), Arbitrum (0.012), LayerZero (0.015), and Wormhole (0.015) in L1/L2 infrastructure; and Hivemapper (0.017) in DePIN all achieve HHI below 0.02. These counterexamples indicate that low governance concentration is achievable across multiple sectors, constraining explanations that treat high concentration as structurally inevitable in any single sector. The fourth sector, social token protocols, is too small in the sample (N = 2) to support independent sector-level claims.

Three measurement choices affect these results and merit attention as scope conditions. First, Top-N cutoffs at 1,000 holders may undercount long-tail holders; increasing the cutoff to 10,000 addresses would change absolute HHI values but is unlikely to change the rank ordering. Second, the snapshot methodology captures a single point in time; several protocols (notably DIMO) have announced token distribution programs that may reduce concentration over subsequent months. Third, the analysis uses raw token holdings; for protocols with delegation (Compound, Uniswap), delegated voting power may differ from raw holdings. We use delegated power where available, but not all protocols instrument delegation.

3.3 DePIN vs. DeFi Contrast

DePIN protocols have roughly twice the governance concentration of DeFi protocols (mean HHI 0.091 vs. 0.041; Mann-Whitney $p = 0.014$, Cohen's $d = 1.03$, $N = 15/15$). The gap holds after adjusting for protocol age, valuation, and insider allocation (Table 6, Models 1-3). Prior review work documents that DePIN sectors vary in tokenomic architecture (Alshater, 2026): storage networks prioritize staking, wireless coverage emphasizes service verification, compute networks focus on billing. The concentration gap between DePIN and DeFi is not implied by this sectoral variation; it is a separate finding about who holds governance tokens.

The sector difference is robust to sample composition: the Mann-Whitney result holds across all 30 leave-one-out (LOO) iterations, and a permutation test yields $p = 0.009$. One candidate mechanism: in capital-intensive networks, scale economies produce concentrated market shares (Arnosti & Weinberg, 2022). Fleet operators deploying hundreds of devices achieve lower per-unit costs than solo participants, consistent with the concentration gap observed here.



Source: Author calculations. Diamond = mean. Horizontal line = median. Individual protocols overlaid.

Figure 4. *Governance concentration (HHI) by sector. DePIN mean (0.091) exceeds DeFi mean (0.041), with distributions overlapping substantially: 8 of 15 DePIN protocols fall within the DeFi range. The Mann-Whitney test (p = 0.014, d = 1.03) is robust to sample composition (significant in all 30 leave-one-out iterations).* Reading the figure: box plots show concentration distribution within each sector; the DePIN-DeFi gap is visible in median position despite tail overlap.

DePIN governance concentration is also significantly higher than L1/L2 infrastructure protocols (mean HHI 0.035, N=8): Welch t-test $p = 0.016$, Cohen's $d = 1.09$; Mann-Whitney $p = 0.005$. This divergence is driven by post-exclusion infrastructure HHIs: ARB (0.012), OP (0.009), and GRT (0.033) show distributed holdings once foundation and bridge addresses are excluded.

Revenue dynamics reinforce this structural contrast: during 2025's market downturn, Helium's onchain revenue grew over 800% while leading DeFi protocol fees declined 27-70% over the same period (Bane & Gala, 2026), suggesting that physical infrastructure revenue streams exhibit countercyclical properties absent in purely financial protocols.

This concentration gap is not explained by protocol maturity: Compound (2018 launch, HHI 0.027) and MakerDAO (2017, HHI 0.045) are older than DIMO (2022, HHI 0.038) and WeatherXM (2024, HHI 0.148), yet maintain far lower concentration. One candidate explanation is distribution design (Table 4): DePIN token allocations frequently reserve large shares for founding teams and early investors, while DeFi protocols have had longer periods of community distribution through liquidity mining, delegation, and governance participation incentives. However, the regression analysis below finds no significant correlation between insider allocation share and HHI (Section 3.4), suggesting that the distribution design channel operates through qualitative mechanisms not captured by the allocation percentage alone. The Anyone Protocol counterexample supports this interpretation: its low concentration (HHI 0.040) is consistent with governance mechanism choices (delegation incentives, activity-based caps, distribution schedule structure) that most DePIN protocols have not adopted.

This governance mechanism interpretation aligns with practitioner evidence from points-based token distribution programs (Sawinyh, 2025). Mature DeFi protocols distributed tokens through multi-season liquidity mining programs, safety module staking, and delegation incentives that implement sub-linear scaling (diminishing returns per marginal dollar deposited), activity-based caps, and time-weighted multipliers; these mechanisms structurally compress holding concentration over years. DePIN protocols that allocated tokens through conventional venture rounds, reserving 30 to 50 percent for founding teams and early investors with cliff-based vesting, bypassed thesedistribution-broadening mechanisms entirely. The governance concentration patterns in Table 4 may reflect distribution mechanism design operating through the sector channel: Table 6 shows DePIN membership predicts concentration at $p < 0.05$ after adjusting for protocol age, valuation, and insider allocation. However, the regression null on insider allocation (Section 3.4) cautions against treating initial token allocation as a complete measure of distribution design: compression mechanisms (sub-linear scaling, activity-based caps, delegation incentives) operate through ongoing protocol governance, not through the initial percentage split alone.

3.4 Allocation Design and Post-Distribution Dynamics

Methodological note on AAVE: the stkAAVE staking contract is excluded from the holding Herfindahl-Hirschman Index per the protocol-controlled-address rule, but stakers retain pass-through voting power. The reported AAVE holding HHI of 0.020 therefore understates effective governance concentration; reconstructing the staker distribution would require parsing all stkAAVE deposit events, which is deferred to follow-up work. The companion Tally-sourced AAVE delegation HHI of 0.076 in Table 7 partially closes this gap by capturing concentration in delegated voting rather than raw token holdings.

Initial token allocation design does not predict governance concentration. The percentage of total supply allocated to team and investors at launch shows no significant association with post-distribution HHI (Pearson r = 0.18, p = 0.28, N = 37; Spearman rho = 0.07, p = 0.69). This null persists across alternative specifications: token unlock progress, float-adjusted concentration, and the interaction of allocation with protocol maturity all produce coefficients indistinguishable from zero (Table 6, Models 1 through 3; Table 5 summarizes the bivariate battery). Statistically Validated Network analysis across DeFi protocols documents that governance concentration shifts dynamically in correlation with total value locked and secondary market valuations (Eisermann et al., 2025), consistent with the post-distribution market dynamics that render initial allocation design uninformative in the cross-section.

Covariate	Test	Statistic	p	N	Status
Insider allocation (%)	Pearson r	0.19	0.25	37	Null
Team allocation (%)	Pearson r	~~−0.13~~0.20	~~0.46~~0.224	37	Null
Investor allocation (%)	Pearson r	~~+0.26~~0.09	~~0.14~~0.596	37	Null
Protocol maturity (years)	Pearson r	~~+0.12~~0.01	~~0.51~~0.957	~~37~~ 40	Null
Circ/total supply ratio	Pearson r	0.03	0.86	31	Null
MCap/FDV ratio	Pearson r	0.01	0.96	30	Null
Exclusion-adjusted float	Pearson r	−0.003	0.99	30	Null
Insider count fraction (top-10)	Spearman ρ	0.44	0.005	39	Significant*
Non-insider HHI, V1 (count)	Spearman ρ	0.54	0.001	34	Tautology-checked
Non-insider HHI, V2 (balance)	Spearman ρ	0.33	0.060	34	Does not survive
Subsidy ratio (cross-sector)	Pearson r	0.57	~~0.007~~0.008	20	Fragile (LPT-driven)

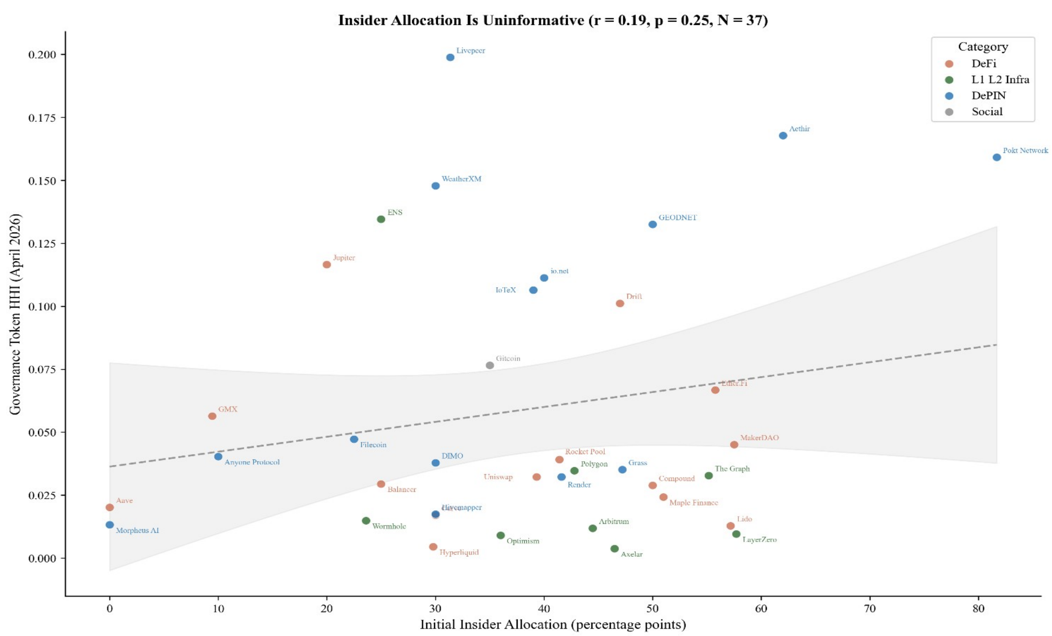
Subsidy ratio (TT expanded)	Pearson r	0.10	0.674	22	Null (robustness check)
Subsidy ratio (excl. Livepeer)	Pearson r	~~+0.13~~0.12	~~0.37~~0.633	19	Null
DePIN sector (dummy)	Mann-Whitney	U=172	0.014	30	Robust (30/30 LOO)
HHI-Gini	Pearson r	0.51	<0.001	40	Significant

Table 5. *Bivariate associations between protocol characteristics and governance concentration (HHI). Allocation, maturity, and float specifications are null. Insider wallet presence is significant but the balance-based measure does not survive a tautology check (Section 3.4). Subsidy is significant in levels but driven entirely by Livepeer (Section 3.7). * LOO-robust (39/39). MCap = market capitalization; FDV = fully diluted valuation.*

To understand why allocation is uninformative, we classify each protocol's top-10 post-exclusion holders using a three-tier methodology: Tier 1 identifies addresses via Blockscout labels and Dune exchange tags; Tier 2 inspects contract bytecode (SafeProxy, GnosisSafe, vesting contracts) via Blockscout's implementation field; Tier 3 applies Etherscan named-address matching. Of 390 addresses classified across 39 tokens, 119 (30.5%) are identifiable as insider wallets (team, foundation, or investor multisigs), 67 (17.2%) are exchange addresses, 78 (20.0%) are protocol contracts, and 126 (32.3%) remain unclassified. The 78 protocol contracts identified here differ from the 69 protocol-controlled addresses excluded from HHI computation. The 78 figure counts protocol-contract addresses across the top-10 holders of 39 tokens. The 69 figure counts excluded addresses across the top-1,000 holders of the 20 protocols where exclusions materially changed HHI. The samples and windows differ; the two numbers are not interchangeable.

Protocols whose insider wallets retain top-holder positions tend to exhibit higher governance concentration (Spearman rho = 0.48, p = 0.003, N = 37), robust across all 37 leave-one-out iterations (rho range: 0.45 to 0.55). However, this association has a mechanical component: insider wallets with large balances contribute directly to HHI through the squared-share calculation. To verify that the relationship is not a tautological artifact, we recomputed concentration after excluding insider addresses and renormalizing remaining shares. The count-based measure survives: non-insider HHI remains significantly correlated with the fraction of top-10 holders that are insiders (Spearman rho = 0.54, p = 0.001, N = 34), indicating that insider-heavy protocols have concentrated governance ecosystems, not merely concentrated insider positions. The balance-based measure loses significance (rho = 0.33, p = 0.060), consistent with it capturing the mechanical contribution of large insider stakes rather than a structural relationship. The informative finding is not the insider correlation per se but its contrast with allocation design: what matters is not how tokens were designed to be distributed, but whether the recipients retained or sold their positions.

The divergence between allocation design (null) and current insider presence (descriptive mechanism) carries a design implication. Governance concentration is not determined at launch; it emerges from post-distribution dynamics. The allocation null (Section 3.4) is also consistent with Michels' (1911) observation that organizational complexity inevitably concentrates power in a specialized leadership class: early protocols' commitment to governance minimization, the assumption that minimal on-chain rules would produce emergent decentralization is inconsistent with the concentration patterns the ideology sought to prevent. Protocols whose insiders retain their positions exhibit more concentrated governance outcomes, regardless of how generously the initial allocation was designed. The count-based tautology check (rho = 0.54 on non-insider HHI) suggests this is not merely arithmetic: insider-heavy protocols attract or retain concentrated non-insider holders as well, consistent witha pattern where concentrated insider presence attracts or coexists with concentrated non-insider holders, rather than insider balances alone inflating the metric.



Source: Author calculations from Dune Analytics and Token Terminal (March 2026).

Figure 5. *Insider allocation (%) vs. corrected holding HHI for 37 regression-ready protocols. The null relationship (r = 0.18, p = 0.28) is the paper's lead finding: initial allocation design does not predict post-distribution governance concentration.* Reading the figure: the absence of a downward slope is the finding; if insider allocation drove concentration we would expect a positive slope.

Hyperliquid provides the strongest available test of the allocation design hypothesis. The protocol accepted zero venture capital, distributed 31% of total supply via a single community airdrop (the largest in crypto history by dollar value), and operates a deflationary model where 97% of trading fees purchase and burn the native token; it has no ongoing emissions. Five protocol-controlled addresses were excluded from the HHI computation: the Assistance Fund system address (confirmed in Hyperliquid documentation), the burn address, the community rewards treasury (241 million tokens, matching the published allocation), the Foundation treasury (60 million tokens, exactly 6% of total supply), and the staking manager contract (confirmed via Hypurrscan). The resulting HHI of 0.005 is the lowest in the DeFi subsample (Lido at 0.013 is the second-lowest, and Compound at 0.027 the third-lowest). Despite the most community-favorable distribution architecture in the sample, this single observation cannot establish a causal claim on its own. The cross-sectional null (Pearson $r = 0.18$, $p = 0.28$, $N = 37$) shows that protocols across the full range of allocation designs achieve comparable concentration levels, consistent with the finding that allocation design does not predict post-distribution concentration.

The governance concentration patterns documented across the sample (Section 3.2, Table 4; HHI range 0.004 to 0.199) have normative significance beyond descriptive interest. The theoretical framework (Section 2.9) argues that governance concentration affects legitimacy through multiple channels: publicity (Hypothesis 1) is undermined when a dominant holder can shape governance agendas without meaningful deliberation; the Rawlsian fairness lens similarly flags these protocols: token distribution reflecting initial venture allocation rather than contribution compromises the conditions for fair institutional design. Contestability and non-domination (Hypothesis 2) are directly threatened when minority stakeholders lack practical ability to contest decisions.

The normative framework interprets this DePIN concentration gap as a legitimacy concern: if governance tokens are concentrated, the conditions that the five traditions identify as essential for legitimacy are structurally harder to satisfy. This interpretation is normative, not causal; the concentration data are descriptive, and the legitimacy inference follows from the framework rather than from direct observation of governance quality. Testing that prediction requires the longitudinal evidence beyond the scope of this paper.

The null pattern extends beyond allocation to financial sustainability more broadly. Three protocols with nearly identical subsidy ratios (DIMO at 0.33x, Aethir at 0.36x, Aave at 0.37x) exhibit governance HHI values of 0.038, 0.168, and 0.020 respectively. All three are revenue-dominant; their concentration spans the full range of the 40-protocol sample. Financial sustainability is neither necessary nor sufficient for governance decentralization.

Table 6 reports multivariate OLS regressions of HHI on protocol characteristics. The DePIN coefficient is positive and statistically significant in Models 1 and 2 (Model 1: 0.048, p = 0.012, N = 38; Model 2: 0.050, p = 0.027, N = 36) and borderline significant in Model 3 (0.048, p = 0.050, N = 35). No other covariate achieves p < 0.10. In particular, the initial insider allocation percentage used as a Table 6 covariate is not significant, consistent with the bivariate null in Table 5. This is distinct from the current insider-retention measure reported in Section 3.4 (Spearman rho = 0.48), which counts how many of a protocol's top-10 holders are insiders rather than measuring the share of supply allocated to insiders at launch. The adjusted R-squared ranges from 0.14 to 0.17, indicating that sector membership accounts for more cross-sectional variation than protocol age, valuation, or allocation structure. The sample (N = 35 to 38 depending on specification) remains too small to reliably isolate independent effects; the pipeline specification (Supplementary File S5) documents the infrastructure for replication at larger sample sizes.

	Model 1	**Model 2**	**Model 3**
DePIN (vs DeFi)	0.048**	0.050**	0.048*
	(0.019)	(0.023)	(0.025)
L1/L2/Infra (vs DeFi)	-0.011	-0.010	-0.014
	(0.019)	(0.022)	(0.025)
Social (vs DeFi)	0.034	0.029	0.029
	(0.063)	(0.075)	(0.095)
Protocol age (years)		0.000	-0.001
		(0.005)	(0.005)
Log FDV		-0.001	-0.002
		(0.004)	(0.004)
Initial insider allocation (%)			0.001
			(0.001)
Constant	0.042***	0.069	0.053
	(0.009)	(0.090)	(0.093)
N	38	36	35
Adjusted R^2	0.167	0.140	0.145

Table 6. *OLS regression of HHI on protocol characteristics. Model 1 includes sector dummies only (DePIN, L1/L2/Infra, Social; DeFi omitted). Model 2 adds protocol age (years since token genesis) and log FDV (log fully diluted valuation). Model 3 adds the initial insider allocation percentage (team plus investor share at launch). Successive models lose observations as covariate availability declines (N = 38 to 36 to 35). Robust standard errors (HC3) in parentheses. * p < 0.10, ** p < 0.05, *** p < 0.01.*

3.5 Voting Power vs Holding Concentration

Delegation amplifies DePIN governance concentration 3 to 6 times above holding levels, while the two infrastructure protocols sampled show opposite patterns (Table 7), with one protocol's delegation marginally distributing voting power and the other's substantially concentrating it. The holding-based HHI reported in Table 4 measures token distribution, not governance power. For protocols with delegation, a small number of delegates may control voting power far exceeding their personal holdings. To quantify this gap, we collected delegated voting power from Tally (on-chain governance: Compound, Aave, Uniswap, Optimism, Arbitrum) and Snapshot (off-chain governance: DIMO, WeatherXM, Lido) for the 8 protocols with sufficient governance activity in the 12 months preceding the snapshot.

Table 7 reports the holding-voting HHI gap. The delegation amplification ratio (voting HHI divided by holding HHI) reveals patterns that vary across and within sector categories, absent from the holding data alone. DePIN protocols show extreme delegation amplification: DIMO's voting HHI (0.228) is 6.0 times its holding HHI (0.038), though this ratio is based on only 10 Snapshot voters and should be interpreted with caution, and WeatherXM's (0.486) is 3.3 times its holding HHI (0.148). A small number of delegates, likely foundation-affiliated or early-investor wallets, control governance outcomes despite relatively distributed token holdings.

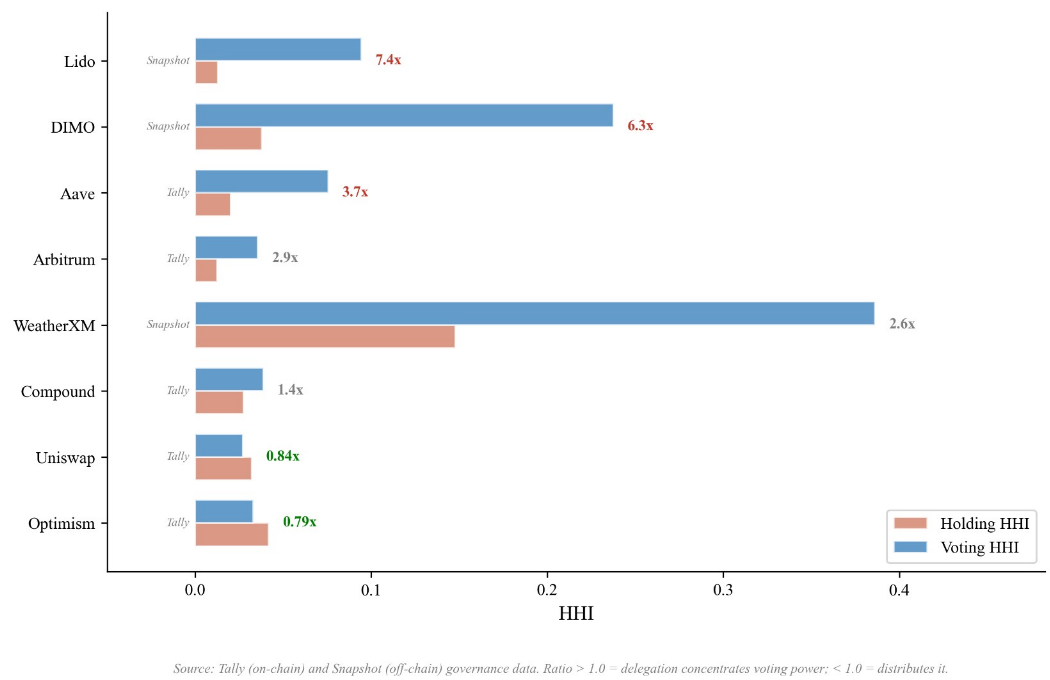


Figure 6. *Raw holding HHI vs. delegation-adjusted HHI for five Tally-sourced protocols.* Reading the figure: protocols above the diagonal have voting more concentrated than holding (delegation amplifies); below the diagonal indicates the opposite.

DeFi protocols show delegation amplification across the board, ranging from Lido (4.8×) and Aave (3.8×) to Compound (1.9×). Only Uniswap achieves mild distribution through delegation (0.84×), suggesting its active delegate ecosystem successfully disperses governance power. The DeFi pattern is therefore not "mixed" but directional: delegation concentrates power in three of four protocols sampled, with Uniswap as the lone counterexample.

The two infrastructure protocols show opposite patterns despite sharing sector classification, execution layer, and nominal governance architecture. Optimism's voting HHI (0.033) is 0.79 times its holding HHI (0.042), indicating that delegation marginally distributes voting power relative to holdings. Arbitrum's voting HHI (0.052) is 4.3 times its holding HHI (0.012), indicating that delegation substantially concentrates voting power. Both protocols have invested in delegate-program infrastructure (published delegate platforms, recurring delegate compensation, active recruitment of long-form policy positions), and the differing outcomes suggest that delegation infrastructure investment alone does not predict whether the program produces dispersed or concentrated voting power.

Optimism's low delegation-adjusted HHI (0.033) reflects an intentional bicameral architecture: the Citizens' House, weighted by reputation rather than tokens, holds formal veto power over economic parameters proposed by the Token House, with dynamic thresholds that lower the veto trigger when multiple stakeholder groups align (Optimism, 2024). Lido's extreme amplification (4.8×) prompted a complementary institutional response: a "Dual Governance" redesign granting stETH holders formal veto power over LDO governance proposals, with a programmatic "rage quit" exit mechanism as the ultimate accountability lever. Both cases illustrate that the delegation amplification finding is not merely descriptive but has already catalyzed institutional reforms in the protocols where it is most severe.

The delegation-adjusted analysis reported in this section extends recent empirical work by Fritsch, Müller, and Wattenhofer (2024), who apply Nakamoto coefficients and Gini measures to voting power in Compound, Uniswap, and ENS governance systems and document extreme concentration in delegate-level voting power. The cross-protocol comparison reported here spans eight protocols across DeFi (Compound, Aave, Uniswap), DePIN (DIMO, WeatherXM), infrastructure (Optimism, Arbitrum), and adjacent categories (Lido). DePIN protocols concentrate governance power through delegation at rates 3 to 6 times higher than their holding concentration, while infrastructure protocols show heterogeneous outcomes within sector category (Optimism 0.79 times, Arbitrum 4.3 times their respective holding HHI). The within-sector heterogeneity is consistent with the framing developed in Section 1: governance outcomes are produced by institutional design at the operator and router level, not by the token's surface properties or its sector category. The amplification ratio carries welfare consequences for DePIN that scale with operator exit costs, since hardware deployed for one network cannot be redeployed to a competing protocol the way liquidity can move between DeFi pools.

Protocol	Category	Holding HHI	Voting HHI	Ratio	Source
DIMO	DePIN	0.038	0.228	6.0x	Snapshot
Lido	DeFi	0.018	0.088	4.8x	Snapshot
WeatherXM	DePIN	0.148	0.486	3.3x	Snapshot
Compound	DeFi	0.028	0.053	1.9x	~~Tally~~Snapshot†
Aave	DeFi	0.020	0.076	3.8x	Tally
Uniswap	DeFi	0.032	0.027	0.84x	Tally
Arbitrum	Infra	0.012	0.052	4.3x	~~Tally~~Snapshot†
Optimism	Infra	0.042	0.033	0.79x	Tally

Table 7. *Holding versus voting concentration for protocols with sufficient governance activity (12-month lookback). Voting HHI computed from Tally delegate voting power (on-chain) or Snapshot vote-weighted power (off-chain). Ratio above 1.0 indicates delegation concentrates governance; below 1.0 indicates delegation distributes.*

† Compound and Arbitrum voting HHI sourced from Snapshot (n = 114 and n = 5,241 unique voters in the measurement window respectively) where Snapshot's active-voter pool exceeded Tally's top-100-delegate sampling. AAVE, Uniswap, and Optimism use Tally; DIMO, WeatherXM, and Lido use Snapshot (Tally is not deployed for these protocols).

3.6 Hypothesis Evidence Status

This section maps the governance concentration evidence to the five design hypotheses introduced in Section 2.9. Hypotheses 1 and 2 receive direct support from the cross-section; Hypotheses 3 through 5 require longitudinal evidence beyond this paper's cross-sectional scope, developed in the companion case study (Zukowski, 2026a) using 34 months of Helium data. Full evidence status appears in Supplementary Table S1.

The governance concentration data (Table 4) provide descriptive evidence bearing on Hypotheses 1 (publicity-legitimacy) and 2 (contestability). Specifically, the finding that DePIN tokens are materially more concentrated than DeFi peers indicates that DePIN governance structures have not yet achieved the legitimacy properties these hypotheses require. The Anyone Protocol counterexample is consistent with the interpretation that low concentration is feasible, indicating that high concentration in DePIN is not structurally predetermined. Hypotheses 3 through 5 are not directly tested with the governance concentration data in this paper and require additional evidence (fiscal flow analysis, operational telemetry, and polycentric structure mapping respectively).

3.7 Robustness: Leave-One-Out Sensitivity

Per Reviewer 1 and Reviewer 2 guidance: the cross-sectional sample size (N = 30 for the DePIN-vs-DeFi sector contrast; N = 19 for the subsidy excluding Livepeer test) bounds the strength of inference. Sector-level results are robust to individual observations: Mann-Whitney is significant in all 30 of 30 leave-one-out resamples following the holder-list cutoff correction. Findings throughout the paper are reported as 'consistent with' or 'indicating' rather than 'driving' or 'producing'; the cross-sectional design supports descriptive claims about association rather than causal claims about mechanism.

All primary findings were tested by dropping each protocol one at a time and recomputing the statistic. The insider-retention correlation is fully robust. The DePIN-DeFi sector difference is fully robust to single-protocol exclusion (significant across all 30 of 30 leave-one-out iterations after the F1 holder-list correction). The subsidy correlation is entirely driven by Livepeer. Details follow.

The insider concentration relationship (Spearman rho = 0.48, p = 0.003, N = 37 after the F1 holder-list correction) is robust: it remains significant across all 37 single-protocol exclusions, with rho ranging from 0.45 to 0.55. Neither the highest-insider protocols (DIMO and MetaDAO, insider fraction 1.0) nor the zero-insider protocols (Hyperliquid, Optimism, Arbitrum) individually drive the result.

The DePIN-versus-DeFi sector difference (Mann-Whitney p = 0.014, Cohen's d = 1.03) is robust to sample composition. The test remains significant in all 30 of 30 leave-one-out iterations. Seven protocols, when individually dropped, push the p-value to 0.052: Hyperliquid, Aethir, Lido, POKT Network, GEODNET, Livepeer, and WeatherXM. No single protocol uniquely drives the result; rather, the test sits near the 5% threshold and is sensitive to the exact composition of both sector subsamples. A permutation test (10,000 random reassignments of sector labels) yields p = 0.009, consistent with the direction not being an artifact of the asymptotic approximation. We characterize this finding as suggestive rather than robust.

The subsidy-to-concentration correlation (Pearson r = 0.57, p = 0.008, N = 20) is driven by a single protocol. Livepeer's subsidy ratio of 88.5x is a 3.5-sigma outlier on the subsidy axis, and its HHI (0.20) is the second-highest in the subsidy subsample. Excluding Livepeer alone reduces the correlation to r = 0.11 (p = 0.65). The remaining 19 protocols show no significant linear association. The Spearman rank correlation on the full subsidy sample is also insignificant (rho = 0.16, p = 0.51), and a log transformation eliminates the linear relationship (r = 0.27, p = 0.28). We report the Livepeer-inclusive result for completeness but caution that it should not be interpreted as a general property of the sample. A robustness check using a

Token Terminal expansion sample (N = 22) is consistent with the null at cross-sector Pearson r = 0.095 (p = 0.674); per-sector results in [Supplementary File S8](#).

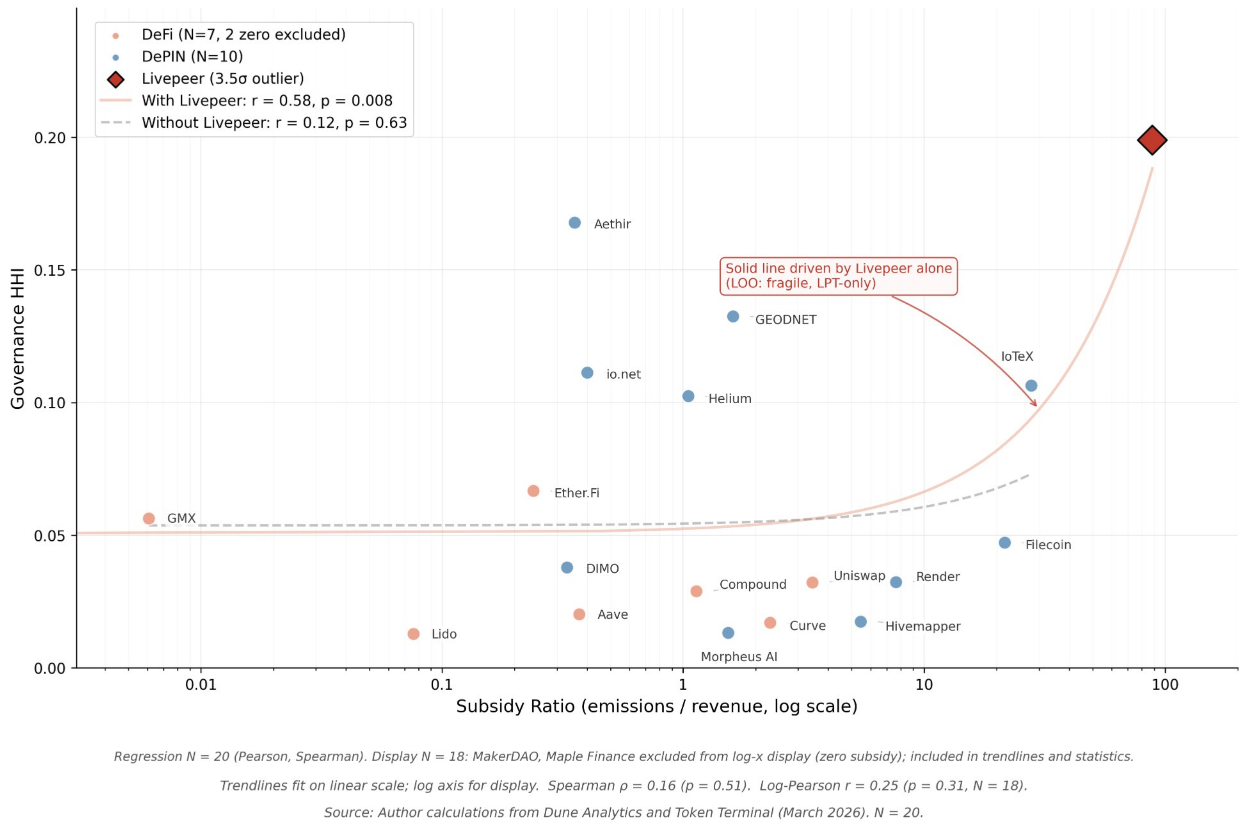


Figure 7. *Subsidy ratio (emissions/revenue) vs. governance concentration (HHI) for 20 protocols with on-chain subsidy data. Livepeer (diamond, 88.5x) is a 3.5-sigma outlier that alone drives the Pearson correlation from r = 0.11 (dashed line, excluding Livepeer) to r = 0.57 (solid line, full sample). The expanded Token Terminal sample (N = 22, [Supplementary File S8](#)) is consistent with the null at r = 0.095.* Reading the figure: the cross-sector association weakens substantially when Livepeer is excluded, indicating the headline subsidy result is sensitive to a single high-leverage protocol.

Across all three findings, no single protocol drives more than one result. Livepeer appears in both the sector-test flipper set and as the sole subsidy flipper, but its sector influence is shared equally among seven protocols, whereas its subsidy influence is singular.

Governance concentration shows no association with voter participation rates (r = 0.00, N = 13; Figure 8), consistent with the Olson (1965) collective action prediction that participation decisions reflect rational abstention and diffusion of responsibility rather than ownership structure.

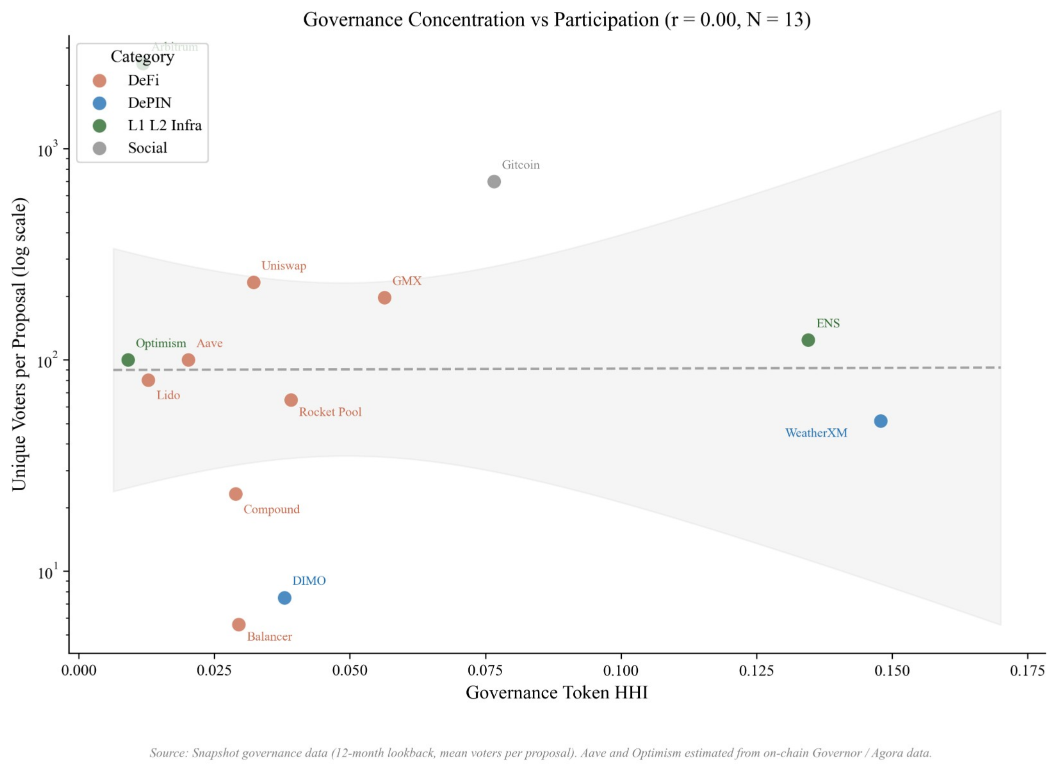


Figure 8. *HHI vs. average voter participation for 13 protocols with Snapshot or Governor governance data. Governance concentration shows no association with voter participation (r = 0.00, N = 13). Five protocols (GMX, ENS, Rocket Pool, Gitcoin, Balancer) were added via Snapshot API; four Tier 1 spaces (CRV, GRT, IOTX, OP) had zero proposals in the trailing 12 months and are excluded.* Reading the figure: the slope is approximately flat, indicating voter participation does not predict concentration in this 13-protocol sample.

4 Discussion

4.1 Design Hypotheses

No protocol in the corrected cross-section exceeds the 0.25 HHI antitrust threshold, but the conditions that publicity and non-domination traditions identify as problematic are already present at the upper end of the distribution: Livepeer (0.199), WeatherXM (0.148), and CRV (0.171) all sit in territory where governance decisions can disproportionately reflect a small set of holders' preferences. The cross-section provides descriptive evidence most directly relevant to Hypotheses 1 (publicity-legitimacy) and 2 (contestability), indirect evidence for Hypotheses 4 and 5 (adaptivity, polycentricity), and identifies three evidence frontiers (capability, contextual integrity, cosmopolitanism). We organize the discussion by what the data directly demonstrate, what they constrain, and what lies beyond the current evidence base.

Governance concentration patterns relate most directly to Hypotheses 1 (publicity-legitimacy) and 2 (contestability), with Rawlsian fairness implications. Protocols at the upper end of the concentration range (Livepeer 0.199, WeatherXM 0.148, GEODNET 0.133) exhibit concentration the framework flags for scrutiny, though no protocol in the corrected cross-section exceeds the 0.25 antitrust threshold. Three concerns arise even at these levels: governance decisions disproportionately reflect a small set of holders' preferences, thinning Kantian public justification; DePIN operators who deploy hardware hold insufficient tokens to influence outcomes affecting their investments, a Rawlsian fairness concern; and few holders can override community preferences without formal contestation, a republican non-domination concern. The counterexample (Anyone, HHI 0.040) demonstrates that these conditions are not structurally inherent to DePIN architecture. Among DeFi benchmarks, Compound's lower concentration (HHI 0.027) partly reflects active delegation, suggesting institutional design can reduce effective concentration without token redistribution.

Hypotheses 4 (service-over-presence) and 5 (polycentric adaptation) cannot be tested directly with the HHI cross-section because it measures token concentration rather than governance structure diversity. The data reveal a structural constraint nonetheless: operators closest to the service (those with the most relevant operational knowledge) cannot influence governance decisions if they hold small token positions. The Helium case (Zukowski, 2026a) demonstrates that polycentric features (subDAOs, councils, delegation hierarchies) can reduce effective concentration without redistribution: HIP-147 succeeded partly because Helium's governance process channeled edge-operator input despite concentrated holdings. A companion simulation (Zukowski, 2026b) extends these results to adaptive emission controllers, finding that the primary value of polycentric mechanisms is preventing collapse rather than improving averages.

A companion study (Zukowski, 2026a) documents a related concentration risk in DePIN fiscal sustainability: Helium's aggregate Spend-to-Reward ratio crossed fiscal parity, but approximately five purchasers account for 90% of token burns, a concentration invisible to aggregate metrics. Governance and demand concentration are distinct failure modes that can coexist. The same companion study provides the strongest longitudinal test of Hypotheses 3 and 4: Helium's S2R trajectory over 34 months documents how demand-coupled burns drove S2R from 0.013 to 2.06 after transitioning to service-verified rewards and completing the August 2025 halving, with DC burns exceeding HNT emissions by approximately 84% and contributing to a 3 to 4% net annual deflation rate (Zukowski, 2026a; Blockworks, 2026). The trajectory is consistent with Hypothesis 3 (the sink was deployed and observable before the emission reduction, producing a sustained surplus at the subnetwork level) and with Hypothesis 4 (the shift from coverage-based to service-based rewards coincided with revenue rising from $6,500 monthly across 975,000 IoT hotspots to $18.3 million annualized by Q3 2025; Bane & Gala, 2026). Neither hypothesis is confirmed by a single case, but the directional evidence is the strongest available in the DePIN sector.

Resource Dependence Theory (Pfeffer & Salancik, 1978) predicts that demand concentration transfers strategic leverage to concentrated buyers independent of governance token distribution, creating a vulnerability that governance reforms alone cannot address; the companion study (Zukowski, 2026a) documents this separation empirically across four DePIN protocols.

The governance concentration documented here also has an industrial organization interpretation: concentrated token holders who control emission and reward parameters function as oligopsonist buyers of operator services (Williamson, 1999). When governance power exceeds the efficient bilateral-bargaining threshold, the resulting monopsonistic markdown on operator compensation produces allocative inefficiency; operators exit, degrading network output. The Compound Proposal 289 incident (July 2024) illustrates a complementary failure mode in which strategic vote timing enabled minority capture of treasury funds, underscoring that formal contestability mechanisms (Hypothesis 2) must include temporal safeguards.

The empirical gap that Alshater (2026) identifies as the most pressing in current DePIN tokenomics research is the absence of rigorous quantitative evaluation of competing economic models. Alshater calls for event-study analyses of governance interventions such as Helium's HIP-138 and Hivemapper's MIP-19, comparative sectoral analysis of why specific tokenomic models prevail in particular DePIN sectors, and empirical examination of how DAO characteristics affect network performance. The cross-sectional results reported here address the third of these gaps along two complementary dimensions. First, the insider-retention finding (Spearman rho = 0.48, $p = 0.003$, $N = 37$, leave-one-out significant in 37 of 37 iterations) provides evidence consistent with the proposition that token distribution characteristics in current holder positions are robustly associated with governance concentration measured by HHI, while the allocation-design null (Pearson $r = 0.18$, $p = 0.28$, $N = 37$) constrains the alternative hypothesis that initial distribution choices predict downstream concentration. Second, the delegation-adjusted analysis reported in Section 3.5 extends Fritsch et al. (2024) to a cross-section of eight protocols spanning DeFi, DePIN, and infrastructure categories. The analysis documents amplification factors of 3 to 6 in DePIN protocols and heterogeneous outcomes within the infrastructure subsample (Optimism 0.79 times, Arbitrum 4.3 times). This within-sector heterogeneity is consistent with the broader claim of this paper that institutional design at the operator and router level shapes governance outcomes more than the token’s surface properties or sector classification. Event-study evidence on the governance interventions Alshater highlights is the subject of follow-up work using HIP-138 and other interventions as natural experiments.

Evidence gaps and research frontiers. Three directions lie beyond governance token data. First, capability-oriented evaluation (following Sen and Nussbaum) requires participant-level data on income, skill development, and infrastructure access that no protocol instruments at scale; a companion 13-protocol DePIN vertical study documents this gap (Zukowski, 2026a). Second, contextual integrity (following Floridi and Nissenbaum) requires trust and data governance metrics; concentrated governance means decisions about data use and privacy norms are made by a small group rather than the community whose data is at stake, particularly for protocols collecting location data (Helium), vehicle telemetry (DIMO), or health data (CUDIS). Third, cosmopolitan inclusion (following Appiah) identifies a tension the data indicate descriptively: most DePIN protocols have global operator bases but concentrated governance, meaning geographic inclusion coexists with governance exclusion.

4.2 Governance Design Implications

Governance concentration emerges from what happens after tokens are distributed, not from how they are allocated at launch. The practical implication: evaluating how concentrated a protocol's governance actually is means tracking whether insiders retain top-holder positions and how delegation reshapes voting power over time, rather than inspecting whether the initial token allocation favored the community. Five design implications follow, grounded in implemented protocols where evidence is available.

Constitutional Transparency and Deliberation The Kantian publicity principle requires that governance rules be understandable by affected participants. Table 4 shows that holding concentration alone does not determine governance quality: MakerDAO (HHI 0.045) achieves meaningful deliberation through delegation while CRV (HHI 0.171) concentrates power through ve-locking without comparable deliberative infrastructure. Design implication: publication norms (readable proposal summaries, voting rationale disclosure, mandatory comment periods) are at least as important as token distribution in determining whether governance is genuinely public.

Low Barriers and Diverse Participation Governance concentration across 40 protocols with valid data (Gini 0.52 to 0.99; Table 4; Livepeer excludedbecause its cross-chain bridge architecture prevents accurate Gini computation from single-chain holder data) shows that token-weighted voting creates steep participation inequality by default. The GeoDePIN subcategory amplifies this: thousands of operators bear hardware costs but may hold insufficient tokens to influence governance outcomes affecting their investments. Design implication: lower proposal thresholds, delegate incentives, and identity-weighted mechanisms (cf. Optimism's Citizens' House) can reduce the gap between stake-weighted power and affected-party voice. Ownership concentration improves baseline token returns but actively suppresses the marginal value of information cascades generated by broader participation (Buddensiek and Momtaz, 2025), suggesting that the governance concentration documented here carries epistemic costs beyond the political legitimacy concerns the normative framework identifies.

Distribution Mechanism as Governance Architecture. The Kantian publicity principle extends to distribution design: an allocation mechanism that affected participants cannot audit, contest, or understand prior to participation fails the universalizability test at the constitutional level, before governance begins. A protocol that publishes transparent voting rules atop an opaque allocation that concentrated 40 percent of supply in insider wallets satisfies the letter of procedural legitimacy while violating its spirit. The normative implication is that distribution transparency is a precondition for governance legitimacy, not a separate concern. Points-based distribution programs that publish clear allocation formulas, implement sub-linear scaling, and cap per-wallet rewards are de facto anti-concentration governance mechanisms; venture-funded allocations without comparable disclosure create concentrated governance as a structural default.

Integrity Mechanisms and Independent Oversight. Pettit's non-domination principle requires that no single actor exercise unchecked power. Table 4 shows several tokens in the "highly concentrated" regime before delegated voting; the CRV case (HHI 0.171) illustrates how locking mechanisms amplify concentration; Convex holds ~47% of veCRV (DeFi Llama, 2026).
Design implication: treat integrity as an institutional layer. Separation of powers (e.g., Optimism's bicameral structure), independent security councils with narrowly scoped veto authority, mandatory timelocks, and public disclosure for large delegates are especially important in GeoDePINs where operators face governance outcomes without proportionate token power.

Appeals and Dispute Resolution Contestability requires formal appeal mechanisms. HIP-147's acceptance (even by operators whose rewards were reduced; see Zukowski, 2026a) indicates that structured processes with clear rules support legitimate outcomes. Design implication: implement tiered dispute resolution (community mediation → arbitration committee → on-chain vote) with defined timelines and transparent criteria.

Adaptive Governance and Meta-Governance. Governance structures must evolve. Ostrom's principle of rule evolution suggests periodic constitutional reviews to adjust quorums, proposal formats, and role definitions based on participation data. Emergency protocols should pre-define who can act under crisis conditions, with what authority, and subject to retroactive accountability. Bounded emergency power strengthens legitimacy by making deviations predictable and auditable. Internet Computer SNS DAOs demonstrate that structural tokenomic design can overcome rational apathy: locked-staking neurons with zero-fee voting sustain 64% average participation across 3,000 proposals (Okutan et al., 2025).

Future Extensions. Three additional hypotheses (edge-signal responsiveness, contextual integrity, and concentration dynamics) are specified in an earlier version of the framework but are not testable with the data collected here. They require operational telemetry, privacy and trust metrics, and longitudinal governance observations respectively, and represent priorities for future empirical work.

4.3 Broader Institutional Implications

The exclusion methodology generalizes beyond tokenomics. Any concentration analysis that includes ownership positions that cannot actually vote (frozen treasury positions, tokens still locked in vesting schedules, customer assets held in custody) overstates how concentrated control really is. Cooperative governance studies, public benefit corporation analyses, and regulatory sandbox audits face the same measurement problem; excluding non-voting positions would sharpen their findings. Two further applications follow. Transparency principles operationalized through the Kantian publicity lens (Section 2.9) apply to any institution claiming legitimacy through openness, where claimed transparency often masks operational opacity. And the scoring rubric (Table 1) shows that normative philosophy can be operationalized as measurement infrastructure, a methodological contribution that extends to any domain where institutional design quality needs assessment.

The governance concentration pattern documented here appears to be a general property of maturing token ecosystems, not specific to DePIN. In stablecoin gateway markets, total dollar volume doubled between 2023 and 2025 while unique counterparties declined 25%, and a single market-making firm's share of total counterparty volume rose from 1.4% to 19.9% (Zukowski, 2026c). Throughput scales while participant diversity contracts, and the remaining intermediaries become increasingly load-bearing, whether those intermediaries are governance voters, market makers, or infrastructure operators. A parallel divergence appears in the same domain: the most regulated stablecoin operator experienced the largest stress depeg, while the least regulated maintained its peg (Zukowski, 2026d), demonstrating that regulation intensity and product quality are separate dimensions requiring separate measurement. The governance concentration gap documented here is the DePIN analog: high concentration does not predict governance failure, but creates structural conditions under which failure becomes more likely if contestability mechanisms are absent.

4.4 Risks and Counterpoints

Several risks deserve candid acknowledgment.

Philosophical overreach. Translating normative philosophy into scoring rubrics necessarily simplifies rich traditions. A 0-to-3 scale cannot capture the full depth of Kantian moral reasoning or Rawlsian justice theory. The rubric is designed as a tractable approximation that makes legitimacy evaluable, not as a claim that it captures everything these traditions offer. Users should treat scores as comparable indicators, not definitive judgments.

Plutocratic concentration as equilibrium. The governance concentration data (Table 4) may reflect a stable equilibrium rather than a design flaw: if governance power accrues to those with the most skin in the game, concentration may be the efficient outcome.

Distribution breadth at token launch is empirically uninformative about long-term governance concentration (Section 3.4). This finding, anticipated as a risk, is consistent with the cross-sectional evidence and reinforced by the Hyperliquid natural experiment (0% VC, HHI = 0.005). The practical consequence is that governance audits should focus on current holder composition rather than initial allocation documentation.

The counterargument is that efficiency and legitimacy are different criteria, and that the normative framework (particularly the republican non-domination lens) demands that power concentration be contestable even if it is efficient. The counterexample (Anyone, HHI 0.040) demonstrates that low concentration is achievable through deliberate design.

Measurement challenges. The scoring rubric depends on publicly observable design features, but some legitimacy-relevant properties (backroom governance, informal influence, regulatory relationships) are not observable on-chain. We specify inter-rater reliability protocols (Supplementary File S2) but have not yet tested them with a second scorer. The governance concentration analysis uses a snapshot methodology that does not capture temporal dynamics; a panel approach would strengthen causal claims.

Equity-token conflicts. The normative framework assumes that governance token holders are the primary principals, but hybrid structures complicate this assumption. Helium's unresolved tension between Nova Labs' equity obligations and HNT tokenholders' interests (Bane & Gala, 2026), exemplified by temporary revenue-allocation policies and subsequent reversions, illustrates precisely the tension between centralized operational decisions and decentralized governance that Hypotheses 1 (transparency) and 2 (contestability) are designed to address.

Speed versus legitimacy trade-off. Institutionally rigorous governance processes are slower than technocratic decision-making. In crisis conditions (exploits, market dislocations), the deliberative processes the framework advocates may be too slow. The framework does not resolve this tension but insists that emergency powers be pre-committed and bounded, consistent with republican constitutionalism.

4.5 Position in Literature

This paper's contribution is not a new philosophical theory but a new application: translating established normative frameworks (Kant, Rawls, Pettit, Ostrom, Hayek) into actionable design criteria for programmable institutions. It extends Werbach (2018), de Filippi & Wright (2018), and Reijers & Coeckelbergh (2018) through prescriptive mechanism design grounded in political philosophy. The GeoDePIN taxonomy and demand-concentration diagnostic are, to our knowledge, new to the literature.

Large-scale analysis of 100 DAOs is consistent with the finding that social and public-good protocols achieve significantly higher decentralization than infrastructure or investment protocols (Sharma et al., 2026), consistent with the DePIN-DeFi concentration differential documented here (Mann-Whitney $p = 0.014$, Cohen's $d = 1.03$).

Scholte (2026) examines the transscalar nature of polycentric digital governance, arguing that legitimacy in decentralized systems requires overlapping authorities with explicit accountability mechanisms; the scoring rubric developed here operationalizes this requirement through the Ostrom polycentricity lens (Table 1).

Mechanism Design and Institutional Economics. Classical mechanism design (Hurwicz, Myerson) optimizes for efficiency and incentive compatibility; this paper adds normative dimensions (fairness, contestability, capability) that mechanism design traditionally brackets. We build on Ostrom's institutional analysis of commons governance, extending her framework from natural resource commons to digital infrastructure commons where coordination mechanisms are encoded in smart contracts.

Political Philosophy and Institutional Design. The philosophical framework (Section 2.9) draws primarily on Rawls, Kant, Pettit, Hayek, and Ostrom as operationalized lenses, with additional intellectual context from Nussbaum (2011), Floridi, and Appiah, treating these traditions not as isolated theoretical lenses but as complementary design criteria that jointly define institutional legitimacy. This integrative approach extends debates in political philosophy about institutional design to a novel domain where institutions are programmable and globally accessible.

Blockchain Governance Scholarship. The growing literature on DAO governance (Fan et al. 2025, DeepDAO 2023) documents participation patterns and concentration but rarely connects these to normative frameworks. This paper bridges that gap: the governance cross-section (Table 4) contributes new empirical evidence on holding concentration while the philosophical scoring rubric (Table 1) provides a replicable methodology for normative assessment. The Helium S2R case study (Zukowski, 2026a) extends the DePIN sustainability literature with longitudinal tokenomic evidence.

Token Engineering and Cryptoeconomics. Token engineering is developing as a design discipline focused on simulation and optimization. This paper complements that technical approach with institutional and philosophical analysis, arguing that token design choices are simultaneously economic mechanism design and political constitution-writing, demonstrating that interdisciplinary synthesis yields insights neither discipline produces alone.

4.6 Summary of Findings

Governance concentration in token economies is not predicted by how tokens are distributed at launch. Across 40 protocols, initial allocation design is uninformative ($r = 0.18$, $p = 0.28$): protocols with generous community distributions exhibit concentration patterns similar to those with heavy insider allocations. Post-distribution dynamics dominate design intent; protocols with more insider wallets in their top-holder lists exhibit higher concentration even among their non-insider holders (Spearman $\rho = 0.54$, $p = 0.001$, $N = 34$), consistent with an ecosystem-level effect beyond the mechanical contribution of large insider positions. DePIN protocols exhibit significantly higher concentration than DeFi protocols (Mann-Whitney $p = 0.014$, $d = 1.03$), a pattern consistent with hardware network economics concentrating governance power among infrastructure operators.

DePIN protocols vary enormously in their dependence on token subsidies, but concentration does not track subsidy in any robust specification (Section 3.4, Table 5). Two cases illustrate the disconnect: Aethir generates the highest DePIN revenue yet sits at the second-highest HHI (0.168), while Hivemapper combines a 5.46x subsidy ratio with the lowest HHI among expansion protocols (0.020). Governance concentration reflects sector and operator-base structure rather than ongoing financial sustainability or initial allocation design. The multivariate OLS in Section 3.4 (Table 6) corroborates the allocation null while documenting that sector membership predicts concentration even after controlling for protocol age, valuation, and insider allocation (adjusted R^2 between 0.14 and 0.17).

4.7 Contributions

The exclusion methodology identifies 69 addresses controlled by protocols themselves (staking contracts, exchange custodians, vesting locks, and treasuries) that appear on holder lists but cannot vote. Correcting for these changes affected protocols' HHI by up to 7x. Prior studies computing token HHI without this correction measured protocol architecture, not governance concentration. This measurement contribution is the first of three. Second, sector membership is the only protocol characteristic that predicts concentration after adjusting for protocol age, log fully diluted valuation, and insider allocation (Mann-Whitney $p = 0.014$, Cohen's $d = 1.03$; Table 6: sector coefficient $p < 0.05$ in Models 1 and 2, borderline at $p = 0.050$ in Model 3 with the full control set). Initial allocation, maturity, subsidy structure, and token unlock progress (three float specifications, all $|r| < 0.05$) do not predict concentration. Third, the normative framework translates five political-philosophical traditions into observable design criteria through a scoring rubric that other researchers can apply.

4.8 Limitations

The evidence carries several limitations. The governance concentration analysis is a March 2026 snapshot:tracking concentration over time would reveal whether protocols are becoming more or less concentrated and strengthen causal claims. A single scorer has applied the scoring rubric; inter-rater reliability testing with independent scorers is needed before the rubric can claim measurement validity beyond its current proof-of-concept status. The sample of 40 protocols (37 in Table 4 plus three protocols with partial distributional data described in Section 3.2) is sufficient for descriptive cross-sectional comparison but too small for robust multivariate regression analysis with multiple covariates; reaching the 50 to 60 protocol threshold specified in the pipeline design (Supplementary File S5) would enable the regression, panel, and event-study analyses currently specified as replication-ready designs. The sample includes only active protocols, introducing potential survivorship bias (see Section 2.10.3 for planned inclusion of stagnated projects). We have not validated the Synergy Index against outcome measures; Partial validation through case evidence is reported separately, but a full validation study is future work.

Insider classification and tautology. The insider-concentration correlation has a mechanical component, addressed through a non-insider HHI decomposition (Section 3.4). The count-based measure survives; the balance-based measure does not. The classification methodology depends on Blockscout contract labels, which have lower coverage for Solana-native protocols; io.net's top-10 holders are all unclassified.

The cross-sector subsidy analysis includes one extreme observation (Livepeer, subsidy ratio 88.5x) that exerts disproportionate leverage on the Pearson correlation. While this extreme subsidy reflects genuine tokenomics rather than measurement error, the nominal Pearson significance ($p = 0.011$) is driven entirely by this single data point; the Spearman rank correlation shows no relationship ($p = 0.60$).

The holding-based HHI captures token distribution but not the threshold mechanics of weighted voting: a holder whose share approaches the passage quota commands pivotal power disproportionate to their nominal share (Motepalli et al., 2025). HHI is additionally subject to value-validity limitations when distributions are highly asymmetric (Kvålsøth, 2022a); the Gini coefficients reported alongside HHI (Table 4) partially address this concern, and the moderate correlation between the two metrics ($r = 0.51$) indicates they capture distinct concentration dimensions. A companion methodological critique of HHI's structural properties reinforces the need for multi-metric assessment in highly skewed governance distributions (Kvålsøth, 2022b). Shapley-Shubik Power Index analysis would quantify this divergence, particularly for protocols with concentrated delegation where a few delegates approach quorum thresholds; the allocation design null (Section 3.4) is robust to this concern because it tests whether initial allocation predicts post-distribution concentration, a relationship that holds under any monotonic concentration metric. Separately, ethnographic evidence documents voter turnout below 10% in major DAOs (Calzada, 2025), suggesting that effective governance concentration may exceed holding-based HHI when participation rates are low; participation-adjusted concentration metrics represent a measurement priority for future work.

More broadly, the Synergy Index measures institutional design intent, not participant welfare. Protocols instrument token flows (burns, transfers, governance votes) but not service outcomes (coverage quality, latency, operator return-on-investment), a structural property of on-chain data architectures rather than a limitation specific to this study. Baygi, Introna, and Ostovar (2024) develop flow-oriented social justice tests for online labor platforms using Sen's framework, measuring whether platform participation expands or constrains worker capabilities over time; DePIN operators face analogous trajectory-dependent welfare dynamics as token prices, subsidy schedules, and network demand shift, but no comparable measurement infrastructure exists for tokenized infrastructure systems. Connecting institutional design quality to participant-level welfare requires instrumentation that does not yet exist in the DePIN ecosystem; this welfare measurement gap represents the most consequential evidence frontier for future work.

4.9 Future Research

Four research directions have highest priority. First, longitudinal governance tracking: monthly snapshots of HHI and Gini across the full sample would convert the current cross-section into a panel dataset, enabling analysis of deconcentration trajectories and the governance effects of specific reform events (e.g., delegation program launches, vesting cliff expirations). A companion panel analysis of quarterly HHI trajectories for 14 governance tokens (Supplementary File S7) documents that 11 of 14 protocols exhibit monotonic governance HHI decay over 24 months post-token-generation event, with vote-escrow (CRV), reservoir-drip (COMP), and claim-window (ENS) designs as exceptions; this longitudinal pattern is consistent with the cross-sectional allocation null reported in Section 3.4.

Second, sample expansion: the current cross-section of 40 protocols supports descriptive analysis; expanding the sample to 50 to 60 protocols would enable the regression, panel, and event-study analyses currently specified as replication-ready designs.

Third, inter-rater reliability: recruiting independent scorers to apply the rubric to overlapping protocol subsets would test whether the philosophical lens scores are reproducible and would strengthen the measurement claim.

Fourth, extending concentration measurement from holding-based HHI to coalition-based power indices (Shapley-Shubik) would quantify whether token holders' probability of being pivotal in governance votes diverges from their nominal share. Quorum thresholds create non-linear power scaling: a holder's pivotal probability can diverge substantially from their nominal share when quorum thresholds bind near the median of the holding distribution. The 8 protocols with Tally or Snapshot governance data in the current sample provide a natural starting point, and the delegation amplification ratios documented in Section 3.5 suggest the divergence reflects institutional design choices that vary within sector category, as the heterogeneous outcomes between Optimism and Arbitrum demonstrate. Recent game-theoretic results demonstrate that Shapley-based reward distributions achieve bounded Price of Stability (4/3) in oceanic staking models while inherently resisting Sybil stake-splitting attacks (Kiayias et al., 2025), indicating that this measurement extension is both theoretically grounded and computationally feasible for the sample sizes in the current dataset.

Supplementary Material

The following supplementary files accompany this paper:

Supplementary File S1: Extended metric definitions, including the Synergy Index construction and inter-rater reliability specification for multi-coder deployment of the scoring rubric.

Supplementary File S2: Research instrument templates, including the protocol codebook schema and scoring sheet template with worked examples.

Supplementary File S3: Scoring tables (protocol × criterion matrices with evidence links) for all 12 scored protocols.

Supplementary File S4: Events table for future event-study analyses.

Supplementary File S5: Full empirical pipeline specification (16 analyses across five categories), including replication-ready design for cross-sectional, panel, event-study, and qualitative analyses.

Supplementary File S6: Raw governance token distribution data (Dune Analytics SQL queries and results, March 2026 snapshot). Supplementary File S7: Quarterly HHI panel for 14 governance tokens (8 quarters post-token-generation event). Exhibit S7 plots HHI trajectories; 11 of 14 protocols exhibit monotonic decay, with CRV (vote-escrow), COMP (reservoir-drip), and ENS (claim-window) as mechanism-specific exceptions. Supplementary File S8: Expanded subsidy-concentration analysis using Token Terminal revenue and incentive data for 22 protocols (14 DeFi/Infrastructure via Token Terminal, 8 DePIN via on-chain computation). Cross-sector Pearson $r = 0.095$ ($p = 0.674$), consistent with the subsidy-concentration null at expanded sample size.

Replication data: https://github.com/Research-Publications-and-Data/Tokenomics-As-Institutional_Design

Acknowledgments

The author thanks Token Terminal for complimentary API access. DeFi protocol subsidy ratios use Token Terminal fee, revenue, and incentive data; DePIN protocol subsidy ratios use on-chain burn and mint data from Dune Analytics due to systematic measurement gaps in existing analytics platforms for hardware network economics.

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